

EEE 111 — Final Exam Solution Guide

Recognize the Type → Pick the Formula → Solve

Covers **Chapter 4 (BJT DC Biasing)**, **Chapter 1 (Diode Equation)**, and **Chapter 2 (Diode Applications)**

Based on the questions in `EEE111_qestions.docx` — all 52 questions solved (28 from Chapter 4, 5 from Chapter 1, 19 from Chapter 2).

How to Use This Guide

This guide follows the **same order as your question file**: Chapter 4 first, then Chapter 1, then Chapter 2.

Study order that works best:

1. Read **Part 1** for one chapter. Learn how to *recognize* each type.
2. Copy the formulas for that chapter from **Part 2** onto one page by hand.
3. Do the questions of that type from **Part 3**. Cover the solution first, try it yourself, then check.
4. The night before the exam, read only the **Final Exam Quick Revision** section.

Constants used throughout:

Symbol	Value	Where used
V_{BE}	0.7 V	Silicon BJT, always
V_K (Si)	0.7 V	Silicon diode turn-on
V_K (Ge)	0.3 V	Germanium diode turn-on
V_K (GaAs)	1.2 V	GaAs diode turn-on
V_K (Ideal)	0 V	Ideal diode
k/q	8.62×10^{-5} V/K	Thermal voltage

Two things flagged in your document (details inside — I did not invent values):

- Fig. 2.155(a): the first digit of the battery is cut off at the image edge. It does not change the answer.
- Ch. 1 Q16 and Q17: the I_s values are written in **mA**, which is very large for a reverse saturation current. Both answers are given.

PART 1 — HOW TO RECOGNIZE AND SOLVE

Chapter 4 — BJT DC Biasing

Golden rule for the whole chapter: every problem is solved by writing **KVL around the base-emitter loop first** to get I_B , then $I_C = \beta I_B$, then **KVL around the collector-emitter loop** to get V_{CE} .

Learn to spot the configuration by **what is connected to the base**.

What you see at the base	Configuration
One resistor R_B going to V_{CC} , emitter grounded	Fixed bias
One resistor R_B to V_{CC} , plus R_E at emitter	Emitter bias
Two resistors R_1 , R_2 forming a divider	Voltage-divider bias
R_F going from base to the collector (not V_{CC})	Collector feedback
Output taken at emitter, no R_C	Emitter follower
Base grounded / base at a fixed voltage, input at emitter	Common base

Type 1: Fixed-Bias — Find the Q-point

How to recognize this type:

- One single resistor R_B from V_{CC} to the base.
- The emitter goes **straight to ground** (no R_E).
- Question asks for I_{BQ} , I_{CQ} , V_{CEQ} , V_C , V_B , V_E .

What I need to find:

- I_B , then I_C , then V_{CE} and the three node voltages.

Given information:

- V_{CC} , R_B , R_C , β .

Formula:

$$\begin{aligned}
 I_B &= (V_{CC} - 0.7) / R_B \\
 I_C &= \beta \times I_B \\
 V_{CE} &= V_{CC} - I_C \times R_C \\
 V_C &= V_{CE} \quad V_E = 0 \text{ V} \quad V_B = 0.7 \text{ V}
 \end{aligned}$$

- V_{CC} = supply voltage (V), R_B = base resistor (Ω), R_C = collector resistor (Ω), β = current gain (no unit).

How to solve:

1. Write $I_B = (V_{CC} - 0.7) / R_B$. Answer will be in **μA** .
2. Multiply by β to get I_C . Answer will be in **mA** .
3. $V_{CE} = V_{CC} - I_C R_C$.
4. Because the emitter is grounded: $V_E = 0$, so $V_C = V_{CE}$ and $V_B = 0.7 \text{ V}$.

Quick exam trick:

> **No $R_E \rightarrow$ fixed bias.** Only R_B matters for I_B . $V_C = V_{CE}$ and $V_B = 0.7 \text{ V}$ for free.

Type 2: Emitter-Bias — Find the Q-point

How to recognize this type:

- One resistor R_B from V_{CC} to base, **and** a resistor R_E between the emitter and ground.
- Words like “emitter-stabilized bias circuit”.

What I need to find:

- $I_B, I_C, I_E, V_E, V_C, V_{CE}, V_B$.

Given information:

- $V_{CC}, R_B, R_C, R_E, \beta$.

Formula:

$$\begin{aligned} I_B &= (V_{CC} - 0.7) / (R_B + (\beta + 1) R_E) && \leftarrow \text{the key formula} \\ I_C &= \beta I_B && I_E = (\beta + 1) I_B \\ V_E &= I_E R_E \\ V_C &= V_{CC} - I_C R_C \\ V_{CE} &= V_C - V_E \\ V_B &= V_E + 0.7 \end{aligned}$$

How to solve:

1. Use the big denominator: $R_B + (\beta + 1)R_E$. **This is the step people forget.**
2. Get I_C and I_E .
3. $V_E = I_E R_E$ (use I_E , not I_C).
4. $V_C = V_{CC} - I_C R_C$ (use I_C , not I_E).
5. $V_{CE} = V_C - V_E$, and $V_B = V_E + 0.7$.

Quick exam trick:

> R_E present $\rightarrow$ the emitter resistor gets multiplied by $(\beta + 1)$ when seen from the base. That single change turns Type 1 into Type 2.

Type 3: Voltage-Divider Bias — Find the Q-point**How to recognize this type:**

- **Two** resistors R_1 (top) and R_2 (bottom) forming a divider at the base.
- Almost always an R_E at the emitter too.

What I need to find:

- Same list: $I_B, I_C, V_{CE}, V_C, V_E, V_B$.

Given information:

- $V_{CC}, R_1, R_2, R_C, R_E, \beta$.

There are two methods. Check the condition first:

$$\text{Condition (Eq. 4.33): } \beta R_E \geq 10 R_2$$

Formula — EXACT (Thévenin) method — always valid:

$$\begin{aligned} R_{Th} &= R_1 R_2 / (R_1 + R_2) && (\text{parallel}) \\ E_{Th} &= V_{CC} \times R_2 / (R_1 + R_2) && (\text{divider}) \\ I_B &= (E_{Th} - 0.7) / (R_{Th} + (\beta + 1) R_E) \\ I_C &= \beta I_B && I_E = (\beta + 1) I_B \\ V_E &= I_E R_E && V_B = V_E + 0.7 \\ V_C &= V_{CC} - I_C R_C \\ V_{CE} &= V_C - V_E \end{aligned}$$

Formula — APPROXIMATE method — only if $\beta R_E \geq 10 R_2$:

$$\begin{aligned} V_B &= V_{CC} \times R_2 / (R_1 + R_2) && (\text{ignore } I_B \text{ completely}) \\ V_E &= V_B - 0.7 \\ I_E &= V_E / R_E && \text{and } I_C \approx I_E \\ V_{CE} &= V_{CC} - I_C (R_C + R_E) \\ I_B &= I_C / \beta \end{aligned}$$

How to solve:

1. Test $\beta R_E \geq 10R_2$. If the question says “use the approximate approach”, just check and state it.
2. If using exact: find R_{Th} and E_{Th} first, then treat it exactly like Type 2.
3. If using approximate: no I_B at all in the divider — read V_B straight off the divider.
4. Finish with V_C , V_E , V_{CE} .

Quick exam trick:

> **Two base resistors → Thévenin.** Approximate = “pretend $I_B = 0$ ”. Exact = “replace the divider with R_{Th} and E_{Th} , then it is emitter bias”.

Type 4: Collector-Feedback (Voltage-Feedback) Bias**How to recognize this type:**

- The base resistor R_F connects the base to the **collector node**, not to V_{CC} .
- Words: “collector-feedback”, “voltage feedback network”.
- If two resistors sit in series in that feedback path with a capacitor to ground between them, **add them** (a capacitor is an open circuit for DC).

What I need to find:

- I_B , I_C , V_C , and sometimes V_E , V_{CE} .

Given information:

- V_{CC} , R_F , R_C , R_E (may be 0), β .

Formula:

$$\begin{aligned} I_B &= (V_{CC} - 0.7) / (R_F + \beta (R_C + R_E)) && \leftarrow \text{note: } \beta, \text{ not } (\beta+1) \\ I_C &= \beta I_B \\ V_C &= V_{CC} - I_C R_C \\ V_E &= I_C R_E \\ V_{CE} &= V_C - V_E \end{aligned}$$

How to solve:

1. Add up any series feedback resistors (capacitors → open).
2. Put $R_C + R_E$ inside the bracket, multiplied by β .
3. Get I_C , then the node voltages.

Quick exam trick:

> **R_F touches the collector → the denominator is $R_F + \beta(R_C + R_E)$.** Compare with emitter bias, where it was $R_B + (\beta+1)R_E$.

Type 5: Emitter-Follower**How to recognize this type:**

- Output is taken at the **emitter**.
- The collector connects **directly to V_{CC}** with no R_C .
- Often a split supply (+V at top, -V at bottom).

What I need to find:

- I_B , I_E , V_E , and sometimes V_{BC} , V_{CE} .

Given information:

- Supply(s), R_B (or R_1 , R_2), R_E , β .

Formula:

Single base resistor with split supply (+V and -V):

$$I_B = (V_{top} + |V_{bottom}| - 0.7) / (R_B + (\beta + 1) R_E)$$

$$I_E = (\beta + 1) I_B$$

$$V_E = -|V_{bottom}| + I_E R_E$$

Divider version: find R_{Th} , E_{Th} first, then

$$I_B = (E_{Th} - 0.7) / (R_{Th} + (\beta + 1) R_E)$$

$$V_E = I_E R_E \quad V_C = V_{CC} \quad V_{CE} = V_C - V_E \quad V_{BC} = V_B - V_C$$

How to solve:

1. Walk the base-emitter loop, adding up **both** supplies if the bottom rail is negative.
2. Same denominator as emitter bias: $R_B + (\beta + 1)R_E$.
3. V_E is measured from the **bottom rail**, so add the negative supply back in.
4. $V_C = V_{CC}$ always (no R_C to drop voltage).

Quick exam trick:

> **No $R_C \rightarrow V_C = V_{CC}$.** With a -6 V rail and +6 V rail, the driving voltage is $6 + 6 - 0.7 = 11.3$ V.

Type 6: Common-Base

How to recognize this type:

- The **base is grounded** or held at a fixed DC voltage.
- The input signal goes into the **emitter**.
- The transistor is often drawn lying on its side.

What I need to find:

- I_E first (not I_B), then I_C , then V_C and V_{CE} .

Given information:

- Emitter supply, R_E , collector supply, R_C , sometimes β or a fixed V_B .

Formula:

$$\begin{aligned} V_E &= V_B - 0.7 & (V_B = 0 \text{ if base grounded}) \\ I_E &= (V_E - V_{EE}) / R_E & (V_{EE} = \text{the emitter-side supply}) \\ I_B &= I_E / (\beta + 1) & I_C = \beta I_B \approx I_E \\ V_C &= V_{CC} - I_C R_C \\ V_{CE} &= V_C - V_E & V_{BC} = V_B - V_C \end{aligned}$$

How to solve:

1. Find V_E from the base voltage: $V_E = V_B - 0.7$.
2. Apply Ohm's law across R_E between V_E and the emitter supply to get I_E .
3. If β is given, split into I_B and I_C . If not, use $I_C \approx I_E$.
4. Get V_C , then subtract.

Quick exam trick:

> **Base grounded $\rightarrow V_E = -0.7$ V.** Then everything comes from Ohm's law. Start at the emitter, not the base.

Type 7: Saturation Current (I_{Csat})

How to recognize this type:

- The words "saturation current" or " I_{Csat} ".
- Usually a one-line question referring back to an earlier figure.

What I need to find:

- The maximum possible collector current, when V_{CE} is forced to 0.

Given information:

- V_{CC} and whichever resistors are in the collector-emitter path.

Formula:

Fixed bias (no R_E):	$I_{Csat} = V_{CC} / R_C$
Any circuit with R_E :	$I_{Csat} = V_{CC} / (R_C + R_E)$

How to solve:

1. Short the transistor (imagine $V_{CE} = 0$).
2. Add the resistances that remain in the collector-to-emitter path.
3. Divide V_{CC} by that sum. **β is not used at all.**

Quick exam trick:

> **Saturation = short the transistor.** Just $V_{CC} \div$ (resistors in the C-E path). Takes 10 seconds — free marks.

Type 8: “Reverse” Problems — Given Voltages, Find Resistors / β / V_{CC} **How to recognize this type:**

- The circuit has **letters instead of numbers** on some resistors (R_B , R_C , R_1).
- Instead, node voltages or currents are printed on the figure ($V_C = 7.6$ V, $I_B = 20$ μ A, $I_E = 4$ mA).
- Wording: “Given the information appearing in Fig. ..., determine ...”.

What I need to find:

- Whatever is missing: R_B , R_C , R_E , R_1 , β , or V_{CC} .

Given information:

- Read every number printed on the figure. Those are your knowns.

Formula (rearranged versions of the normal ones):

$R_C = (V_{CC} - V_C) / I_C$	$R_E = V_E / I_E$
$R_B = (V_{CC} - V_B) / I_B$	$\beta = I_C / I_B$
$I_C = I_E - I_B$	$V_B = V_E + 0.7$
$V_{CC} = V_{CE} + I_C R_C + V_E$	
For a divider: $I_2 = V_B / R_2$, $I_1 = I_2 + I_B$, $R_1 = (V_{CC} - V_B) / I_1$	

How to solve:

1. List every value printed on the figure.
2. Find a resistor that has **both** its end voltages and its current known — solve that one first.
3. Use $V_B = V_E + 0.7$ to unlock the base side.
4. Keep going; each answer unlocks the next.

Quick exam trick:

> **Same formulas, just rearranged.** Always start from a resistor where you know voltage across it *and* current through it.

>

> **Careful with R_1 :** the current through R_1 is $I_2 + I_B$, not just I_2 .

Type 9: Design Problems**How to recognize this type:**

- The word “**Design**” or “**Determine R_C and R_B for...**”.
- No figure at all — everything is in the text.
- Says “Use standard values”.

What I need to find:

- Resistor values that produce the requested Q-point.

Given information:

- V_{CC} , β , and the target I_{CQ} and V_{CEQ} .

Formula:

$$\begin{aligned}
 R_C &= (V_{CC} - V_{CEQ} - V_E) / I_{CQ} & (V_E = 0 \text{ if no } R_E) \\
 R_E &= V_E / I_E \\
 I_B &= I_{CQ} / \beta \\
 R_B &= (V_{CC} - 0.7 - V_E) / I_B \\
 \text{Divider design: } R_2 &\leq \beta R_E / 10 & \text{ then } R_1 = V_{CC} R_2 / (V_B - R_2)
 \end{aligned}$$

How to solve:

1. Draw the circuit yourself from the description.
2. Work out every node voltage from the target Q-point.
3. Ohm's law on each resistor.
4. Round to the **nearest standard value** at the end (see the standard-value list in Part 2).

Quick exam trick:

> **Design = normal analysis run backwards.** Find the voltage across each resistor, divide by the current through it, then round.

Type 10: Current-Source Circuits**How to recognize this type:**

- Question says “**Calculate the current through the load**” or “calculate the current I ”.
- The current arrow I is drawn in the collector branch.
- May include a **Zener diode** setting the base voltage.

What I need to find:

- I , which is just I_C .

Given information:

- Supplies, base network, R_E , β .

Formula:

Find V_B from whatever sets it:

- Fixed source through R_B → use KVL with $(\beta+1)R_E$
- Resistor divider → use R_{Th} / E_{Th}
- Zener → $V_B = V_{bottom} + V_Z$ (Zener fixes it directly)

Then:

$$\begin{aligned}
 V_E &= V_B - 0.7 \\
 I_E &= (V_E - V_{bottom}) / R_E \\
 I &= I_C = \alpha I_E = \beta / (\beta+1) \times I_E
 \end{aligned}$$
How to solve:

1. Get V_B . **This is the whole problem.**
2. Subtract 0.7 to get V_E .
3. Ohm's law across R_E gives I_E .
4. I_C is I_E shaved by $\alpha = \beta / (\beta+1)$ — usually a 1% change.

Quick exam trick:

> **The load resistor value does not matter.** That is the whole point of a current source. If a question gives you a “2.2 k Ω load”, it is a distractor — the current is set by the base and R_E .

Chapter 1 — Semiconductor Diodes (The Diode Equation)

All five questions in this chapter use one equation and one constant. There are no circuits.

Type 1: Find the Thermal Voltage V_T

How to recognize this type:

- “Determine the thermal voltage for a diode at a temperature of ...”.
- A temperature in °C is given.

What I need to find:

- V_T in millivolts.

Given information:

- Temperature in °C.

Formula:

$$\begin{aligned} T_K &= T_C + 273 \\ V_T &= k T_K / q \quad \text{where } k/q = 8.62 \times 10^{-5} \text{ V/K} \\ V_T &= (8.62 \times 10^{-5}) \times T_K \end{aligned}$$

- k = Boltzmann constant = 1.38×10^{-23} J/K, $q = 1.6 \times 10^{-19}$ C.

How to solve:

1. Convert °C to Kelvin by **adding 273**.
2. Multiply by 8.62×10^{-5} .
3. Express in mV.

Quick exam trick:

> **Add 273, multiply by 8.62×10^{-5}** . At room temperature (25 °C / 298 K) you should get ≈ 26 mV — use that as a sanity check.

Type 2: Find the Diode Current I_D (forward bias)

How to recognize this type:

- Gives I_s , n , and a **positive** V_D .
- Says “using Eq. 1.2” or “using the diode equation”.

What I need to find:

- I_D in mA.

Given information:

- I_s (reverse saturation current), n (ideality factor, 1 or 2), V_D , temperature.

Formula:

$$I_D = I_s (e^{(V_D / (n V_T))} - 1) \quad \leftarrow \text{Shockley equation (Eq. 1.2)}$$

- n = ideality factor (1 for high current, 2 for low current), V_T = thermal voltage (V).

How to solve:

1. Get V_T for the given temperature (Type 1).
2. Compute the exponent: $V_D \div (n \times V_T)$.
3. Take e to that power on your calculator.
4. Subtract 1, multiply by I_s .

Quick exam trick:

> **n multiplies V_T , not V_D .** Writing V_D/V_T and forgetting n doubles the exponent and destroys the answer.

Type 3: Diode Current under Reverse Bias

How to recognize this type:

- The applied voltage is **negative** (“reverse-bias potential of -10 V ”).
- Often followed by “Is the result expected? Why?”.

What I need to find:

- I_D , which will simply be $-I_S$.

Given information:

- I_S , n, negative V_D .

Formula:

$$I_D = I_S (e^{(V_D / (n V_T))} - 1)$$

For V_D negative and large: $e^{(\text{negative big})} \approx 0$

$$\text{So } I_D \approx I_S (0 - 1) = -I_S$$

How to solve:

1. Note the exponent is a large negative number.
2. $e^{(\text{large negative})} \approx 0$.
3. So $I_D = -I_S$. Answer is the saturation current, flowing backwards.
4. For “is it expected?” — yes, a reverse-biased diode passes only its tiny leakage current.

Quick exam trick:

> **Reverse bias → the answer is just $-I_S$.** No calculator needed.

Type 4: Find I_S (given I_D and V_D)

How to recognize this type:

- “Given a diode current of ... find I_S ”.

What I need to find:

- I_S , usually in pA or nA.

Given information:

- I_D , V_D , n, temperature.

Formula:

$$I_S = I_D / (e^{(V_D / (n V_T))} - 1)$$

How to solve:

1. Find V_T from the temperature.
2. Compute the exponential term.
3. Divide I_D by (that term $- 1$).

Quick exam trick:

> **Same equation, divide instead of multiply.** Expect a very tiny answer (pA/nA) — if you get mA, you inverted it.

Type 5: Find V_D (given I_D and I_S)

How to recognize this type:

- “find the applied voltage V_D ”.
- V_T is often given directly (e.g. 26 mV) so you skip Type 1.

What I need to find:

- V_D in volts — should land near 0.4-0.8 V.

Given information:

- I_D , I_s , n , V_T .

Formula:

$$V_D = n V_T \times \ln (I_D / I_s + 1) \quad \leftarrow \text{natural log, not log}_{10}$$

How to solve:

1. Divide I_D by I_s (watch the powers of ten — this is a huge number).
2. Add 1, take **ln**.
3. Multiply by $n \times V_T$.

Quick exam trick:

> **Rearranged Shockley = ln form.** Use **ln**, never log. Sanity check: the answer must be a sensible diode voltage, roughly 0.3-0.8 V.

Chapter 2 — Diode Applications

Golden rule for the whole chapter: every DC diode problem is answered by first deciding **ON or OFF** for each diode.

The universal test:

1. **Assume the diode is OFF.** Remove it (open circuit) and find the voltage that would appear across it.
2. If the **anode is more positive** than the cathode by more than V_K , it is really **ON**.
3. Redo the circuit with the diode replaced by a battery of V_K (or a wire if ideal).
4. Check your assumption: the current must flow in the arrow's direction. If it comes out negative, the diode is OFF.

Reading the symbol: current flows in the direction the **triangle points**. The **bar** is the cathode (the negative/blocking side).

Type 1: Series Diode Circuit — Find I , V_o , I_D

How to recognize this type:

- One loop with one or two diodes and resistors.
- "Determine the current I " / "Determine V_o and I_D ".
- Diodes labelled Si, Ge, GaAs, or Ideal.

What I need to find:

- Whether each diode conducts, then the loop current and output voltage.

Given information:

- Supply voltages and their polarity, resistor values, diode material.

Formula:

$$I = (\text{Total driving voltage} - \sum \text{diode drops}) / (\sum \text{resistances})$$

$$V_o = I \times R_{\text{output}} \quad (\text{or } V_o = 0 \text{ if no current flows})$$

$$\text{Si} \rightarrow 0.7 \text{ V} \quad \text{Ge} \rightarrow 0.3 \text{ V} \quad \text{GaAs} \rightarrow 1.2 \text{ V} \quad \text{Ideal} \rightarrow 0 \text{ V}$$

How to solve:

1. Mark the polarity of every source. **A battery with "-" on top makes that node negative.**
2. Check each diode's direction against the driving voltage.
3. If OFF $\rightarrow I = 0$ and $V_o = 0$ (no current, no drop across the resistor).
4. If ON $\rightarrow$ subtract every diode drop from the driving voltage, divide by the total resistance.

Quick exam trick:

> **Diode OFF $\rightarrow I = 0 \rightarrow V_o = 0 \text{ V}$.** Half of these questions are answered in one line. Always check the direction before doing any arithmetic.

Type 2: Two Diodes Pointing at Each Other (Back-to-Back)

How to recognize this type:

- Two diodes in the **same branch** with the triangles pointing in **opposite** directions.
- Or one bar-then-triangle and one triangle-then-bar in series.

What I need to find:

- Nothing complicated — that branch is dead.

Given information:

- Just look at the two symbols.

Formula:

$$\text{Opposing diodes in series} \rightarrow \text{branch is an OPEN CIRCUIT} \rightarrow I = 0$$

How to solve:

1. Spot the opposing pair.
2. Delete that whole branch from the circuit.
3. Solve whatever is left (usually a trivial resistor problem).

Quick exam trick:

> **Arrows facing each other = wire cut.** One diode always blocks, whichever way current tries to go.

Type 3: Parallel Diode Branches (both may conduct)**How to recognize this type:**

- Two branches, each with its own diode, joining at the same output node.
- Different diode materials (e.g. one Si, one GaAs) in the two branches.

What I need to find:

- V_o at the common node, then the current in each branch.

Given information:

- Supply, both branch resistances, both diode types.

Formula:

Assume both ON, write KCL at the output node:
 $(V_{\text{supply}} - V_{K1} - V_o)/R_1 + (V_{\text{supply}} - V_{K2} - V_o)/R_2 = V_o / R_{\text{load}}$
 Solve for V_o , then check each branch current is positive.

How to solve:

1. Assume both diodes conduct.
2. KCL at the output node: current in = current out.
3. Solve the single linear equation for V_o .
4. **Check both branch currents are positive.** If one is negative, that diode is OFF — redo with only the other.

Quick exam trick:

> **Two diode branches meeting → KCL at the joint.** Do not assume the smaller-drop diode “wins” — with a resistor in each branch, both usually conduct.

Type 4: Half-Wave Rectifier**How to recognize this type:**

- One diode, sinusoidal input, “half-wave rectifier”.
- Asks you to **sketch** v_i , v_d , i_d , v_L .

What I need to find:

- Peak value V_m , the DC level, and the three waveforms.

Given information:

- Either V_m or the DC level, the frequency, and R .

Formula:

Ideal diode:	$V_{dc} = 0.318 V_m$	$(0.318 = 1/\pi)$
Silicon diode:	$V_{dc} = 0.318 (V_m - 0.7)$	
Peak output:	$V_o(\text{peak}) = V_m - V_K$	
Peak current:	$I_{\text{peak}} = (V_m - V_K) / R$	
Period:	$T = 1 / f$	
PIV rating:	$PIV \geq V_m$	

How to solve:

1. If the DC level is given, work **backwards**: $V_m = V_{dc} / 0.318$ (then add V_K if silicon).

2. Positive half-cycle: diode ON $\rightarrow$ output follows the input (minus V_K), $v_d = V_K$.
3. Negative half-cycle: diode OFF $\rightarrow$ output = 0, and the **whole input appears across the diode**: $v_d = v_i$.
4. i_d has the same shape as the output voltage, scaled by $1/R$.

Quick exam trick:

> **0.318 for half-wave, 0.636 for full-wave.** When the diode is OFF, all the input voltage sits across the **diode**, so v_d reaches $-V_m$. That is the part students draw wrong.

Type 5: Full-Wave Bridge Rectifier**How to recognize this type:**

- Four diodes drawn in a **diamond**.
- Asks for v_o , PIV, and maximum diode current.

What I need to find:

- Output peak, V_{dc} , PIV of each diode, peak diode current.

Given information:

- Input peak, load resistance, diode type.

Formula:

```
V_o(peak) = V_m - 2 V_K      (2 diodes conduct at once; ideal  $\rightarrow V_m$ )
V_dc = 0.636  $\times$  V_o(peak)
PIV (bridge) = V_m
I_max = V_o(peak) / R_L
```

How to solve:

1. Two diodes conduct on each half-cycle, so subtract **two** diode drops (zero if ideal).
2. $V_{dc} = 0.636 \times$ peak output.
3. Each diode blocks V_m when off $\rightarrow$ PIV = V_m .
4. Peak diode current = peak load current.

Quick exam trick:

> **Bridge = two drops, PIV = V_m , factor 0.636.** (For a centre-tapped transformer instead, PIV = $2V_m$ — different circuit, do not mix them up.)

Type 6: Clippers (the output is CUT OFF at a level)**How to recognize this type:**

- Diode is in **parallel** with the output (shunt), or in series with a battery.
- The question says “determine v_o ” and shows a square or sine input.
- The output waveform keeps its shape but has a flat top or flat bottom.

What I need to find:

- The clipping level, and the value of v_o in both states.

Given information:

- Input peak values, R , diode type, DC battery value **and its polarity**.

Formula:

```
Clipping level = V_battery + V_K      (walk from the output node to ground
                                       through the branch, adding as you go)

Diode ON  $\rightarrow v_o$  is held at the clipping level
Diode OFF  $\rightarrow$  no current in that branch  $\rightarrow v_o$  follows the input
              (series clipper:  $v_o = 0$  ; shunt clipper:  $v_o = v_i$ )
```

How to solve:

1. Find the **clipping level**: start at the output node, walk down the diode branch to ground, adding V_K and the battery voltage with correct signs.
2. Decide when the diode turns on — compare v_i against that level.
3. Write v_o for both states.
4. Sketch: flat line at the clipping level for one state, the input shape for the other.

Quick exam trick:

> **Clipper = output is CUT.** Diode in parallel with the output → shunt clipper. Diode in series with the signal → series clipper. Battery pushes the clipping level up or down by its own voltage.

Type 7: Clampers (the output is SHIFTED up or down)**How to recognize this type:**

- There is a **capacitor in series** with the input. **This is the giveaway.**
- Diode and R are in parallel at the output.
- The output waveform has the **same peak-to-peak swing** as the input, just moved.

What I need to find:

- The capacitor voltage V_C , then both output levels.

Given information:

- Input square/sine peaks, C, R, diode type, DC battery.

Formula:

```

Step 1 – the half-cycle where the DIODE CONDUCTS:
  v_o = clamping level      (= V_battery ± V_K, same walk as a clipper)
  V_C = v_i - v_o           ← capacitor charges to this and HOLDS it
Step 2 – the other half-cycle (diode OFF):
  v_o = v_i - V_C
Check: peak-to-peak of v_o must EQUAL peak-to-peak of v_i

```

How to solve:

1. Identify which half-cycle turns the diode ON.
2. In that half, v_o = the clamping level. Compute $V_C = v_i - v_o$.
3. In the other half, the diode is off: $v_o = v_i - V_C$ (V_C is unchanged).
4. **Always check the swing matches the input swing.** If not, you made a sign error.

Quick exam trick:

> **Capacitor in series → clamper → same swing, shifted.** The peak-to-peak check catches almost every mistake.

Type 8: Clamper Design**How to recognize this type:**

- “Design a clamper to perform the function indicated in Fig. ...”.
- Shows an input waveform and the **desired** output waveform side by side.

What I need to find:

- The DC source value, the diode direction, and the R-C condition.

Given information:

- Both waveforms, and whether the diodes are ideal or silicon.

Formula:

```

Shift = v_o(level) - v_i(level)      (must be the same for both levels)
Clamping level = the output level that is HELD FLAT
Required DC source: V_battery = clamping level ∓ V_K
Design rule: 5τ = 5RC ≫ T/2

```

How to solve:

1. Compare the two waveforms — confirm the peak-to-peak values match.
2. Decide **which output level is the clamped one** (it is the one the diode holds).
3. Work out the DC source needed to sit at that level, remembering the V_K offset.
4. Point the diode so it conducts during that half-cycle.
5. State $5RC \gg T/2$.

Quick exam trick:

> **Clamped level** → tells you the battery. **Direction of the shift** → tells you which way the diode points. Shifting up = clamp the bottom; shifting down = clamp the top.

Type 9: R-C Time Constant Check**How to recognize this type:**

- “Calculate 5τ ” and “compare 5τ to half the period”.

What I need to find:

- 5τ , $T/2$, and a comment.

Given information:

- R , C , and the frequency.

Formula:

$$\begin{array}{ll} \tau = R C & 5\tau = 5 R C \\ T = 1 / f & \text{half period} = T / 2 \\ \text{Good clamping if } 5\tau \gg T/2 \end{array}$$

How to solve:

1. Multiply $R \times C$ (watch units: $k\Omega \times \mu F = ms$).
2. $T = 1/f$, halve it.
3. Compare and say by what factor.

Quick exam trick:

> **$k\Omega \times \mu F = \text{milliseconds}$** . $56 k\Omega \times 0.1 \mu F = 5.6 ms$. That shortcut saves time and prevents power-of-ten errors.

Type 10: Zener Regulator — Fixed V_i , Variable R_L **How to recognize this type:**

- A Zener diode drawn in **parallel with the load**, R_S in series with the supply.
- Asks for V_L , I_L , I_Z , I_R for given R_L values.

What I need to find:

- First: **is the Zener ON or OFF?** Then the currents.

Given information:

- V_i , R_S , V_Z , P_{Zmax} , R_L .

Formula:

TEST FIRST – remove the Zener and compute:

$$V = V_i \times R_L / (R_S + R_L)$$

If $V < V_Z \rightarrow$ Zener OFF: $V_L = V$, $I_Z = 0$, $I_L = I_R = V / R_L$

If $V \geq V_Z \rightarrow$ Zener ON:

$$V_L = V_Z$$

$$I_R = (V_i - V_Z) / R_S$$

$$I_L = V_Z / R_L$$

$$I_Z = I_R - I_L$$

Max power: $I_{Zmax} = P_{Zmax} / V_Z$

Min load: $R_{Lmin} = R_S V_Z / (V_i - V_Z)$

How to solve:

1. **Always do the removal test first.** Never assume the Zener is on.
2. If off, it is just a voltage divider.
3. If on, V_L is pinned at V_Z ; find I_R and I_L , subtract for I_Z .
4. For “maximum power”, set $I_Z = P_{Zmax}/V_Z$ and work back to R_L .

Quick exam trick:

> **Test before you trust.** Compute the open-circuit divider voltage first. Also remember I_R is **constant** whenever the Zener is on — only I_L and I_Z trade places.

Type 11: Zener Regulator — Find the Range of V_i

How to recognize this type:

- R_L is **fixed**, and the question asks for the “range of V_i ”.

What I need to find:

- V_{imin} and V_{imax} .

Given information:

- R_S , V_Z , P_{Zmax} , fixed R_L .

Formula:

$I_L = V_Z / R_L$	(fixed, because R_L is fixed)
$I_{Zmax} = P_{Zmax} / V_Z$	
$V_i(min) = V_Z + I_L R_S$	← Zener just barely on, $I_Z = 0$
$V_i(max) = V_Z + (I_L + I_{Zmax}) R_S$	← Zener at full power

How to solve:

1. I_L is fixed — compute it once.
2. Minimum: the Zener is about to turn on, so $I_Z = 0$ and $I_R = I_L$.
3. Maximum: I_Z is at its rated limit, so $I_R = I_L + I_{Zmax}$.
4. Both times: $V_i = V_Z + I_R R_S$.

Quick exam trick:

> **Two extremes: $I_Z = 0$ and $I_Z = I_{Zmax}$.** Same formula, two numbers.

Type 12: Zener Design

How to recognize this type:

- “Design the network ... to maintain V_L at ... for a load variation from 0 mA to ...”.

What I need to find:

- V_Z , R_S , and P_{Zmax} .

Given information:

- V_i , the required V_L , and the load current range.

Formula:

$V_Z = V_L$	(the Zener sets the output)
$R_S = (V_i - V_Z) / I_L(\text{max})$	(size R_S for the heaviest load)
Worst case for the Zener is $I_L = 0$:	
$P_{Z\text{max}} = V_Z \times I_L(\text{max})$	

How to solve:

1. V_Z is simply the required output voltage.
2. Size R_S at the **maximum** load current (that is when the Zener has least to spare).
3. The Zener's worst case is the **minimum** load ($I_L = 0$), when it must absorb all of I_R .
4. $P = V_Z \times$ that current.

Quick exam trick:

> **R_S is sized at max load; the Zener's power is rated at min load.** Opposite extremes — that is the whole trick.

PART 2 — COMPLETE FORMULA SHEET

Only the formulas needed for the questions in your document.

Chapter 4 — BJT DC Biasing

Universal BJT relations (true in every configuration)

Formula	Symbols	Unit	When to use
$I_E = I_C + I_B$	terminal currents	A	Always true
$I_C = \beta I_B$	β = current gain	A	Active region
$I_E = (\beta + 1) I_B$	—	A	Active region
$\alpha = \beta / (\beta + 1)$	$\alpha \approx 0.99$	—	Common base, current sources
$I_C = \alpha I_E$	—	A	When I_E is known first
$V_{BE} = 0.7 \text{ V}$	base-emitter drop	V	Silicon, always
$V_B = V_E + 0.7$	node voltages	V	Links base and emitter sides
$V_{CE} = V_C - V_E$	—	V	Always

Fixed bias

Formula	Unit	When to use
$I_B = (V_{CC} - 0.7) / R_B$	A	Emitter grounded, single R_B
$V_{CE} = V_{CC} - I_C R_C$	V	Collector loop
$I_{C\text{sat}} = V_{CC} / R_C$	A	Saturation, no R_E

Emitter bias

Formula	Unit	When to use
$I_B = (V_{CC} - 0.7) / [R_B + (\beta + 1)R_E]$	A	R_B to V_{CC} plus R_E
$V_E = I_E R_E$	V	Emitter node
$V_C = V_{CC} - I_C R_C$	V	Collector node
$I_{C\text{sat}} = V_{CC} / (R_C + R_E)$	A	Saturation with R_E

Voltage-divider bias

Formula	Unit	When to use
$R_{Th} = R_1 R_2 / (R_1 + R_2)$	Ω	Exact method
$E_{Th} = V_{CC} R_2 / (R_1 + R_2)$	V	Exact method
$I_B = (E_{Th} - 0.7) / [R_{Th} + (\beta + 1)R_E]$	A	Exact method
$\beta R_E \geq 10 R_2$	—	Eq. 4.33 — test before approximating
$V_B = V_{CC} R_2 / (R_1 + R_2)$	V	Approximate method
$V_E = V_B - 0.7, I_E = V_E / R_E$	V, A	Approximate method
$V_{CE} = V_{CC} - I_C (R_C + R_E)$	V	Approximate method
$I_2 = V_B / R_2, I_1 = I_2 + I_B$	A	Reverse problems — finding R_1

Collector-feedback bias

Formula	Unit	When to use
$I_B = (V_{CC} - 0.7) / [R_F + \beta(R_C + R_E)]$	A	R_F from base to collector
$V_C = V_{CC} - I_C R_C$	V	Collector node
$I_B = (V_C - V_B) / R_F$	A	Reverse problems

Common base

Formula	Unit	When to use
$V_E = V_B - 0.7$	V	Base grounded or at a fixed voltage
$I_E = (V_E - V_{EE}) / R_E$	A	Emitter loop
$V_{BC} = V_B - V_C$	V	When asked

Design

Formula	Unit	When to use
$R_C = (V_{CC} - V_{CEQ} - V_E) / I_{CQ}$	Ω	Sizing R_C
$R_E = V_E / I_E$	Ω	Sizing R_E
$R_B = (V_{CC} - 0.7 - V_E) / I_B$	Ω	Sizing R_B
$R_2 \leq \beta R_E / 10$	Ω	Choosing R_2 in a divider design
$R_1 = V_{CC} R_2 / V_B - R_2$	Ω	Choosing R_1

Standard resistor values (E24, $\times 10^n$):

1.0, 1.1, 1.2, 1.3, 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3, 3.6, 3.9, 4.3, 4.7, 5.1, 5.6, 6.2, 6.8, 7.5, 8.2, 9.1

Chapter 1 — Diode Equation

Formula	Symbols	Unit	When to use
$T_K = T_C + 273$	temperature	K	Always first
$V_T = kT_K/q = (8.62 \times 10^{-5})T_K$	thermal voltage	V	Every diode-equation problem
$I_D = I_S(e^{V_D/(nV_T)} - 1)$	Eq. 1.2 , Shockley	A	Find current from voltage
$I_S = I_D / (e^{V_D/(nV_T)} - 1)$	saturation current	A	Find I_S
$V_D = nV_T \ln(I_D/I_S + 1)$	—	V	Find voltage from current
$I_D \approx -I_S$	—	A	Reverse bias (large negative V_D)

- n = ideality factor: **1** at high current, **2** at low current. It multiplies V_T .
- At 25 °C, $V_T \approx 25.7$ mV (commonly rounded to 26 mV).

Chapter 2 — Diode Applications

Diode drops

Material	V_K
Ideal	0 V
Germanium (Ge)	0.3 V
Silicon (Si)	0.7 V
Gallium Arsenide (GaAs)	1.2 V

DC diode circuits

Formula	Unit	When to use
$I = (\Sigma V - \Sigma V_K) / \Sigma R$	A	Series loop, all diodes ON
$I = 0, V_o = 0$	—	Any diode reverse-biased
Opposing diodes $\rightarrow$ open circuit	—	Back-to-back pair
KCL at the output node	—	Parallel diode branches

Rectifiers

Formula	Unit	When to use
$V_{dc} = 0.318V_m$	V	Half-wave, ideal
$V_{dc} = 0.318(V_m - 0.7)$	V	Half-wave, silicon
$V_{dc} = 0.636V_m$	V	Full-wave (bridge)
$V_{o(peak)} = V_m - 2V_K$	V	Bridge (two diodes conduct)
$PIV = V_m$	V	Half-wave and bridge
$I_{max} = V_{o(peak)} / R_L$	A	Peak diode current
$T = 1/f$	s	Period

Clippers and clampers

Formula	Unit	When to use
Clipping level = $V_{battery} + V_K$	V	Clipper — walk the diode branch
$V_C = v_i - v_o$ (during conduction)	V	Clamper step 1
$v_o = v_i - V_C$ (diode off)	V	Clamper step 2
$v_o(p-p) = v_i(p-p)$	V	Clamper check
$5\tau = 5RC \gg T/2$	s	Clamper design rule

Zener regulators

Formula	Unit	When to use
$V = V_i R_L / (R_S + R_L)$	V	Test — Zener removed
$V_L = V_Z$	V	Zener ON
$I_R = (V_i - V_Z) / R_S$	A	Zener ON
$I_L = V_Z / R_L$	A	Zener ON
$I_Z = I_R - I_L$	A	Zener ON
$I_{Zmax} = P_{Zmax} / V_Z$	A	Power limit
$R_{Lmin} = R_S V_Z / (V_i - V_Z)$	Ω	Smallest load that keeps Zener on
$V_{i(min)} = V_Z + I_L R_S$	V	Range of V_i
$V_{i(max)} = V_Z + (I_L + I_{Zmax}) R_S$	V	Range of V_i
$P_Z = V_Z I_Z$	W	Power check

PART 3 — ALL QUESTIONS SOLVED

Questions appear in the same order as your document: **Chapter 4 → Chapter 1 → Chapter 2.**

Chapter 4 — BJT DC Biasing

Fixed-Bias Configuration

Question 1 — Fig. 4.118

Problem Type: Fixed Bias → Find the Q-point (Type 1)

Circuit values read from Fig. 4.118: $V_{CC} = 16\text{ V}$, $R_B = 510\text{ k}\Omega$, $R_C = 1.8\text{ k}\Omega$, $\beta = 120$

Given:

- $V_{CC} = 16\text{ V}$, $R_B = 510\text{ k}\Omega$, $R_C = 1.8\text{ k}\Omega$, $\beta = 120$
- Emitter connected directly to ground

Find:

- I_{BQ} , I_{CQ} , V_{CEQ} , V_C , V_B , V_E

Formula:

$$I_{BQ} = (V_{CC} - 0.7)/R_B ; \quad I_{CQ} = \beta I_{BQ} ; \quad V_{CEQ} = V_{CC} - I_{CQ} R_C$$

Solution:

1. $I_{BQ} = (16 - 0.7) / 510\text{ k}\Omega = 15.3 / 510\text{ 000} = \mathbf{30\text{ }\mu\text{A}}$
2. $I_{CQ} = 120 \times 30\text{ }\mu\text{A} = \mathbf{3.6\text{ mA}}$
3. $V_{CEQ} = 16 - (3.6\text{ mA})(1.8\text{ k}\Omega) = 16 - 6.48 = \mathbf{9.52\text{ V}}$
4. Emitter grounded → $V_E = 0\text{ V}$
5. $V_C = V_{CEQ} + V_E = \mathbf{9.52\text{ V}}$
6. $V_B = V_E + 0.7 = \mathbf{0.7\text{ V}}$

Answer:

- **(a)** $I_{BQ} = 30\text{ }\mu\text{A}$ **(b)** $I_{CQ} = 3.6\text{ mA}$ **(c)** $V_{CEQ} = 9.52\text{ V}$
- **(d)** $V_C = 9.52\text{ V}$ **(e)** $V_B = 0.7\text{ V}$ **(f)** $V_E = 0\text{ V}$

Question 2 — Fig. 4.119

Problem Type: Fixed Bias → Reverse problem (Type 8)

Circuit values read from Fig. 4.119: $V_{CC} = 12\text{ V}$, $I_B = 40\text{ }\mu\text{A}$, $\beta = 80$, $V_C = 6\text{ V}$; R_B and R_C unknown

Given:

- $V_{CC} = 12\text{ V}$, $I_B = 40\text{ }\mu\text{A}$, $\beta = 80$, $V_C = 6\text{ V}$, emitter grounded

Find:

- I_C , R_C , R_B , V_{CE}

Formula:

$$I_{CQ} = \beta I_{BQ} ; \quad R_C = (V_{CC} - V_C)/I_{CQ} ; \quad R_B = (V_{CC} - 0.7)/I_{BQ} ; \quad V_{CEQ} = V_C - V_E$$

Solution:

1. $I_C = 80 \times 40\text{ }\mu\text{A} = \mathbf{3.2\text{ mA}}$
2. Voltage across $R_C = 12 - 6 = 6\text{ V}$
 $R_C = 6 / 3.2\text{ mA} = \mathbf{1.875\text{ k}\Omega}$

$$3. R_B = (12 - 0.7) / 40 \mu\text{A} = 11.3 / 40 \times 10^{-6} = \mathbf{282.5 \text{ k}\Omega}$$

$$4. \text{Emitter grounded} \rightarrow V_E = 0, \text{ so } V_{CE} = V_C = \mathbf{6 \text{ V}}$$

Answer:

- (a) $I_C = 3.2 \text{ mA}$ (b) $R_C = 1.875 \text{ k}\Omega$ (c) $R_B = 282.5 \text{ k}\Omega$ (d) $V_{CE} = 6 \text{ V}$

Question 3 — Fig. 4.120

Problem Type: Fixed Bias → Reverse problem (Type 8)

Circuit values read from Fig. 4.120: $R_C = 2.2 \text{ k}\Omega$, $I_B = 20 \mu\text{A}$, $I_E = 4 \text{ mA}$, $V_{CE} = 7.2 \text{ V}$; V_{CC} , β , R_B unknown

Given:

- $R_C = 2.2 \text{ k}\Omega$, $I_B = 20 \mu\text{A}$, $I_E = 4 \text{ mA}$, $V_{CE} = 7.2 \text{ V}$, emitter grounded

Find:

- I_C , V_{CC} , β , R_B

Formula:

$$I_C = I_E - I_B ; \quad V_{CC} = V_{CE} + I_C R_C ; \quad \beta = I_C / I_B ; \quad R_B = (V_{CC} - 0.7) / I_B$$

Solution:

1. $I_C = I_E - I_B = 4 \text{ mA} - 0.02 \text{ mA} = \mathbf{3.98 \text{ mA}}$
2. Drop across $R_C = (3.98 \text{ mA})(2.2 \text{ k}\Omega) = 8.756 \text{ V}$
 $V_{CC} = 7.2 + 8.756 = \mathbf{15.96 \text{ V}}$
3. $\beta = I_C / I_B = 3.98 \text{ mA} / 20 \mu\text{A} = \mathbf{199}$
4. $R_B = (15.96 - 0.7) / 20 \mu\text{A} = 15.26 / 20 \times 10^{-6} = \mathbf{762.8 \text{ k}\Omega}$

Answer:

- (a) $I_C = 3.98 \text{ mA}$ (b) $V_{CC} = 15.96 \text{ V}$ (c) $\beta = 199$ (d) $R_B = 762.8 \text{ k}\Omega$

Question 4 — Fig. 4.118

Problem Type: Saturation current, fixed bias (Type 7)

Given:

- $V_{CC} = 16 \text{ V}$, $R_C = 1.8 \text{ k}\Omega$ (no R_E)

Find:

- $I_{C\text{sat}}$

Formula:

$$I_{C\text{sat}} = V_{CC} / R_C$$

Solution:

1. Short the transistor ($V_{CE} = 0$). Only R_C limits the current.
2. $I_{C\text{sat}} = 16 / 1800 = \mathbf{8.89 \text{ mA}}$

Answer:

- $I_{C\text{sat}} = 8.89 \text{ mA}$
- *Note:* I_{CQ} from Q1 was 3.6 mA , which is well under 8.89 mA — the transistor is properly in the active region.

Emitter-Bias Configuration

Question 8 — Fig. 4.122

Problem Type: Emitter Bias → Find the Q-point (Type 2)

Circuit values read from Fig. 4.122: $V_{CC} = 20\text{ V}$, $R_B = 270\text{ k}\Omega$, $R_C = 470\text{ }\Omega$, $R_E = 2.2\text{ k}\Omega$, $\beta = 125$

Given:

- $V_{CC} = 20\text{ V}$, $R_B = 270\text{ k}\Omega$, $R_C = 470\text{ }\Omega$, $R_E = 2.2\text{ k}\Omega$, $\beta = 125$

Find:

- I_{BQ} , I_{CQ} , V_{CEQ} , V_C , V_B , V_E

Formula:

$$I_{BQ} = (V_{CC} - 0.7) / [R_B + (\beta + 1)R_E]$$

Solution:

1. Denominator: $(\beta + 1)R_E = 126 \times 2200 = 277\,200\text{ }\Omega$
 $R_B + (\beta + 1)R_E = 270\,000 + 277\,200 = 547\,200\text{ }\Omega$
2. $I_{BQ} = (20 - 0.7) / 547\,200 = 19.3 / 547\,200 = \mathbf{35.27\text{ }\mu A}$
3. $I_{CQ} = 125 \times 35.27\text{ }\mu A = \mathbf{4.41\text{ mA}}$
4. $I_E = 126 \times 35.27\text{ }\mu A = 4.44\text{ mA}$
5. $V_E = I_E R_E = (4.44\text{ mA})(2.2\text{ k}\Omega) = \mathbf{9.78\text{ V}}$
6. $V_C = 20 - (4.41\text{ mA})(470\text{ }\Omega) = 20 - 2.07 = \mathbf{17.93\text{ V}}$
7. $V_{CEQ} = V_C - V_E = 17.93 - 9.78 = \mathbf{8.15\text{ V}}$
8. $V_B = V_E + 0.7 = \mathbf{10.48\text{ V}}$

Answer:

- (a) $I_{BQ} = 35.27\text{ }\mu A$ (b) $I_{CQ} = 4.41\text{ mA}$ (c) $V_{CEQ} = 8.15\text{ V}$
- (d) $V_C = 17.93\text{ V}$ (e) $V_B = 10.48\text{ V}$ (f) $V_E = 9.78\text{ V}$

Question 12 — Fig. 4.122

Problem Type: Saturation current with R_E (Type 7)

Given:

- $V_{CC} = 20\text{ V}$, $R_C = 470\text{ }\Omega$, $R_E = 2.2\text{ k}\Omega$

Find:

- I_{Csat}

Formula:

$$I_{Csat} = V_{CC} / (R_C + R_E)$$

Solution:

1. $R_C + R_E = 470 + 2200 = 2670\text{ }\Omega$
2. $I_{Csat} = 20 / 2670 = \mathbf{7.49\text{ mA}}$

Answer:

- $I_{Csat} = 7.49\text{ mA}$ (compare with $I_{CQ} = 4.41\text{ mA}$ from Q8 — active region confirmed)

Question 10 — Fig. 4.123

Problem Type: Emitter Bias → Reverse problem (Type 8)

Circuit values read from Fig. 4.123: $V_{CC} = 12\text{ V}$, $I_C = 2\text{ mA}$, $V_C = 7.6\text{ V}$, $V_E = 2.4\text{ V}$, $\beta = 80$; R_B , R_C , R_E unknown

Given:

- $V_{CC} = 12\text{ V}$, $I_C = 2\text{ mA}$, $V_C = 7.6\text{ V}$, $V_E = 2.4\text{ V}$, $\beta = 80$

Find:

- R_C , R_E , R_B , V_{CE} , V_B

Formula:

$$R_C = (V_{CC} - V_C) / I_C ; \quad I_E = I_C(\beta+1) / \beta ; \quad R_E = V_E / I_E \\ V_B = V_E + 0.7 ; \quad I_B = I_C / \beta ; \quad R_B = (V_{CC} - V_B) / I_B$$

Solution:

1. Drop across $R_C = 12 - 7.6 = 4.4 \text{ V}$
 $R_C = 4.4 / 2 \text{ mA} = \mathbf{2.2 \text{ k}\Omega}$
2. $I_E = I_C(\beta+1) / \beta = 2 \text{ mA} \times 81/80 = 2.025 \text{ mA}$
 $R_E = 2.4 / 2.025 \text{ mA} = \mathbf{1.185 \text{ k}\Omega}$
3. $V_B = 2.4 + 0.7 = \mathbf{3.1 \text{ V}}$
4. $I_B = 2 \text{ mA} / 80 = 25 \text{ }\mu\text{A}$
 $R_B = (12 - 3.1) / 25 \text{ }\mu\text{A} = 8.9 / 25 \times 10^{-6} = \mathbf{356 \text{ k}\Omega}$
5. $V_{CE} = V_C - V_E = 7.6 - 2.4 = \mathbf{5.2 \text{ V}}$

Answer:

- (a) $R_C = 2.2 \text{ k}\Omega$ (b) $R_E = 1.185 \text{ k}\Omega$ (c) $R_B = 356 \text{ k}\Omega$
- (d) $V_{CE} = 5.2 \text{ V}$ (e) $V_B = 3.1 \text{ V}$

Question 11 — Fig. 4.124

Problem Type: Emitter Bias → Reverse problem (Type 8)

Circuit values read from Fig. 4.124: $I_B = 20 \text{ }\mu\text{A}$, $R_C = 2.7 \text{ k}\Omega$, $V_{CE} = 7.3 \text{ V}$, $V_E = 2.1 \text{ V}$, $R_E = 0.68 \text{ k}\Omega$; β , V_{CC} , R_B unknown

Given:

- $I_B = 20 \text{ }\mu\text{A}$, $R_C = 2.7 \text{ k}\Omega$, $V_{CE} = 7.3 \text{ V}$, $V_E = 2.1 \text{ V}$, $R_E = 0.68 \text{ k}\Omega$

Find:

- β , V_{CC} , R_B

Formula:

$$I_E = V_E / R_E ; \quad I_C = I_E - I_B ; \quad \beta = I_C / I_B \\ V_{CC} = V_{CE} + I_C R_C + V_E ; \quad R_B = (V_{CC} - V_B) / I_B$$

Solution:

1. $I_E = 2.1 / 680 = 3.088 \text{ mA}$
2. $I_C = 3.088 - 0.02 = 3.068 \text{ mA}$
3. $\beta = 3.068 \text{ mA} / 20 \text{ }\mu\text{A} = \mathbf{153.4}$
4. Drop across $R_C = (3.068 \text{ mA})(2.7 \text{ k}\Omega) = 8.28 \text{ V}$
 $V_{CC} = 7.3 + 8.28 + 2.1 = \mathbf{17.68 \text{ V}}$
5. $V_B = V_E + 0.7 = 2.8 \text{ V}$
 $R_B = (17.68 - 2.8) / 20 \text{ }\mu\text{A} = 14.88 / 20 \times 10^{-6} = \mathbf{744 \text{ k}\Omega}$

Answer:

- (a) $\beta = 153.4$ (b) $V_{CC} = 17.68 \text{ V}$ (c) $R_B = 744 \text{ k}\Omega$

Voltage-Divider Bias Configuration

Question 15 — Fig. 4.125

Problem Type: Voltage-Divider Bias → Exact (Thévenin) method (Type 3)

Circuit values read from Fig. 4.125: $V_{CC} = 16 \text{ V}$, $R_1 = 62 \text{ k}\Omega$, $R_2 = 9.1 \text{ k}\Omega$, $R_C = 3.9 \text{ k}\Omega$, $R_E = 0.68 \text{ k}\Omega$, $\beta = 80$

Given:

- $V_{CC} = 16 \text{ V}$, $R_1 = 62 \text{ k}\Omega$, $R_2 = 9.1 \text{ k}\Omega$, $R_C = 3.9 \text{ k}\Omega$, $R_E = 0.68 \text{ k}\Omega$, $\beta = 80$

Find:

- I_{BQ} , I_{CQ} , V_{CEQ} , V_C , V_E , V_B

Formula:

$$R_{Th} = R_1 R_2 / (R_1 + R_2) ; \quad E_{Th} = V_{CC} R_2 / (R_1 + R_2) \\ I_B = (E_{Th} - 0.7) / [R_{Th} + (\beta + 1)R_E]$$

Solution:

1. $R_{Th} = (62 \times 9.1) / (62 + 9.1) \text{ k}\Omega = 564.2 / 71.1 = \mathbf{7.94 \text{ k}\Omega}$
2. $E_{Th} = 16 \times 9.1 / 71.1 = 145.6 / 71.1 = \mathbf{2.048 \text{ V}}$
3. Denominator = $7940 + 81(680) = 7940 + 55\,080 = 63\,020 \Omega$
4. $I_{BQ} = (2.048 - 0.7) / 63\,020 = 1.348 / 63\,020 = \mathbf{21.39 \mu A}$
5. $I_{CQ} = 80 \times 21.39 \mu A = \mathbf{1.71 \text{ mA}}$
6. $I_E = 81 \times 21.39 \mu A = 1.733 \text{ mA}$
 $V_E = (1.733 \text{ mA})(680 \Omega) = \mathbf{1.18 \text{ V}}$
7. $V_C = 16 - (1.71 \text{ mA})(3.9 \text{ k}\Omega) = 16 - 6.67 = \mathbf{9.33 \text{ V}}$
8. $V_{CEQ} = 9.33 - 1.18 = \mathbf{8.15 \text{ V}}$
9. $V_B = 1.18 + 0.7 = \mathbf{1.88 \text{ V}}$

Answer:

- (a) $I_{BQ} = 21.39 \mu A$ (b) $I_{CQ} = 1.71 \text{ mA}$ (c) $V_{CEQ} = 8.15 \text{ V}$
- (d) $V_C = 9.33 \text{ V}$ (e) $V_E = 1.18 \text{ V}$ (f) $V_B = 1.88 \text{ V}$

Question 17 — Fig. 4.126

Problem Type: Voltage-Divider Bias → Reverse problem (Type 8)

Circuit values read from Fig. 4.126: $V_{CC} = 18 \text{ V}$, $R_C = 4.7 \text{ k}\Omega$, $V_C = 12 \text{ V}$, $R_2 = 5.6 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$; R_1 unknown

Given:

- $V_{CC} = 18 \text{ V}$, $R_C = 4.7 \text{ k}\Omega$, $V_C = 12 \text{ V}$, $R_2 = 5.6 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$
- β is not given, so use $I_E \approx I_C$

Find:

- I_C , V_E , V_B , R_1

Formula:

$$I_C = (V_{CC} - V_C) / R_C ; \quad V_E \approx I_C R_E ; \quad V_B = V_E + 0.7 \\ V_B = V_{CC} R_2 / (R_1 + R_2) \rightarrow R_1 = V_{CC} R_2 / V_B - R_2$$

Solution:

1. Drop across $R_C = 18 - 12 = 6 \text{ V}$
 $I_C = 6 / 4.7 \text{ k}\Omega = \mathbf{1.277 \text{ mA}}$
2. $V_E = (1.277 \text{ mA})(1.2 \text{ k}\Omega) = \mathbf{1.53 \text{ V}}$
3. $V_B = 1.53 + 0.7 = \mathbf{2.23 \text{ V}}$
4. From the divider: $R_1 + R_2 = V_{CC} R_2 / V_B = (18 \times 5.6 \text{ k}\Omega) / 2.23 = 100.8 / 2.23 = 45.16 \text{ k}\Omega$
 $R_1 = 45.16 - 5.6 = \mathbf{39.56 \text{ k}\Omega}$

Answer:

- (a) $I_C = 1.277 \text{ mA}$ (b) $V_E = 1.53 \text{ V}$ (c) $V_B = 2.23 \text{ V}$ (d) $R_1 = 39.56 \text{ k}\Omega$

Question 19 — Fig. 4.125**Problem Type:** Saturation current with R_E (Type 7)**Given:**

- $V_{CC} = 16 \text{ V}$, $R_C = 3.9 \text{ k}\Omega$, $R_E = 0.68 \text{ k}\Omega$

Find:

- $I_{C\text{sat}}$

Formula:

$$I_{C\text{sat}} = V_{CC} / (R_C + R_E)$$

Solution:

1. $R_C + R_E = 3900 + 680 = 4580 \Omega$
2. $I_{C\text{sat}} = 16 / 4580 = \mathbf{3.49 \text{ mA}}$

Answer:

- $I_{C\text{sat}} = 3.49 \text{ mA}$ (I_{CQ} was 1.71 mA — active region)

Question 18 — Fig. 4.127**Problem Type:** Voltage-Divider Bias → Reverse problem (Type 8)**Circuit values read from Fig. 4.127:** $R_C = 2.7 \text{ k}\Omega$, $V_C = 10.6 \text{ V}$, $I_B = 20 \mu\text{A}$, $\beta = 100$, $R_2 = 8.2 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$; V_{CC} and R_1 unknown**Given:**

- $R_C = 2.7 \text{ k}\Omega$, $V_C = 10.6 \text{ V}$, $I_B = 20 \mu\text{A}$, $\beta = 100$, $R_2 = 8.2 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$

Find:

- I_C , V_E , V_{CC} , V_{CE} , V_B , R_1

Formula:

$$\begin{aligned} I_C &= \beta I_B ; \quad I_E = (\beta + 1) I_B ; \quad V_E = I_E R_E ; \quad V_{CC} = V_C + I_C R_C \\ I_2 &= V_B / R_2 ; \quad I_1 = I_2 + I_B ; \quad R_1 = (V_{CC} - V_B) / I_1 \end{aligned}$$

Solution:

1. $I_C = 100 \times 20 \mu\text{A} = \mathbf{2 \text{ mA}}$
2. $I_E = 101 \times 20 \mu\text{A} = 2.02 \text{ mA}$
 $V_E = (2.02 \text{ mA})(1.2 \text{ k}\Omega) = \mathbf{2.42 \text{ V}}$
3. $V_{CC} = V_C + I_C R_C = 10.6 + (2 \text{ mA})(2.7 \text{ k}\Omega) = 10.6 + 5.4 = \mathbf{16 \text{ V}}$
4. $V_{CE} = V_C - V_E = 10.6 - 2.42 = \mathbf{8.18 \text{ V}}$
5. $V_B = V_E + 0.7 = \mathbf{3.12 \text{ V}}$
6. Current through R_2 : $I_2 = 3.12 / 8.2 \text{ k}\Omega = 0.381 \text{ mA}$
Current through R_1 : $I_1 = I_2 + I_B = 0.381 + 0.020 = 0.401 \text{ mA}$
 $R_1 = (16 - 3.12) / 0.401 \text{ mA} = 12.88 / 0.401 \text{ mA} = \mathbf{32.11 \text{ k}\Omega}$

Answer:

- (a) $I_C = 2 \text{ mA}$ (b) $V_E = 2.42 \text{ V}$ (c) $V_{CC} = 16 \text{ V}$
- (d) $V_{CE} = 8.18 \text{ V}$ (e) $V_B = 3.12 \text{ V}$ (f) $R_1 = 32.11 \text{ k}\Omega$

Watch out: in step 6, I_1 must include I_B . Using $I_1 = I_2$ alone gives 33.8 k Ω , which is wrong.

Question 21 — Fig. 4.128 (approximate approach)

Problem Type: Voltage-Divider Bias → Approximate method (Type 3)

Circuit values read from Fig. 4.128: $V_{CC} = 18 \text{ V}$, $R_1 = 39 \text{ k}\Omega$, $R_2 = 8.2 \text{ k}\Omega$, $R_C = 3.3 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$, $\beta = 120$

Given:

- $V_{CC} = 18 \text{ V}$, $R_1 = 39 \text{ k}\Omega$, $R_2 = 8.2 \text{ k}\Omega$, $R_C = 3.3 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$, $\beta = 120$

Find:

- First check Eq. (4.33), then I_C , V_{CE} , I_B , V_E , V_B

Formula:

Check: $\beta R_E \geq 10 R_2$
 $V_B = V_{CC} R_2 / (R_1 + R_2)$; $V_E = V_B - 0.7$; $I_E = V_E / R_E \approx I_C$
 $V_{CE} = V_{CC} - I_C(R_C + R_E)$; $I_B = I_C / \beta$

Solution:

- Check Eq. (4.33):** $\beta R_E = 120 \times 1000 = 120\,000 \Omega$; $10R_2 = 10 \times 8200 = 82\,000 \Omega$
 $120 \text{ k}\Omega \geq 82 \text{ k}\Omega \rightarrow$ **condition satisfied**, approximate approach is allowed.
- $V_B = 18 \times 8.2 / (39 + 8.2) = 147.6 / 47.2 = \mathbf{3.13 \text{ V}}$
- $V_E = 3.13 - 0.7 = \mathbf{2.43 \text{ V}}$
- $I_E = 2.43 / 1000 = 2.427 \text{ mA}$, and $I_C \approx I_E = \mathbf{2.43 \text{ mA}}$
- $V_{CE} = 18 - (2.427 \text{ mA})(3.3 \text{ k}\Omega + 1 \text{ k}\Omega) = 18 - (2.427 \text{ mA})(4.3 \text{ k}\Omega) = 18 - 10.44 = \mathbf{7.56 \text{ V}}$
- $I_B = 2.427 \text{ mA} / 120 = \mathbf{20.23 \mu A}$

Answer:

- (a)** $I_C = 2.43 \text{ mA}$ **(b)** $V_{CE} = 7.56 \text{ V}$ **(c)** $I_B = 20.23 \mu A$
- (d)** $V_E = 2.43 \text{ V}$ **(e)** $V_B = 3.13 \text{ V}$

Question 22 — Fig. 4.128 (exact / Thévenin approach + comparison)

Problem Type: Voltage-Divider Bias → Exact method, then compare (Type 3)

Given:

- Same circuit as Question 21

Find:

- The same five quantities using Thévenin, then compare with Q21

Formula:

$R_{Th} = R_1 R_2 / (R_1 + R_2)$; $E_{Th} = V_{CC} R_2 / (R_1 + R_2)$
 $I_B = (E_{Th} - 0.7) / [R_{Th} + (\beta + 1)R_E]$

Solution:

- $R_{Th} = (39 \times 8.2) / (39 + 8.2) = 319.8 / 47.2 = \mathbf{6.78 \text{ k}\Omega}$
- $E_{Th} = 18 \times 8.2 / 47.2 = \mathbf{3.13 \text{ V}}$ (same as V_B in Q21)
- Denominator = $6775 + 121(1000) = 6775 + 121\,000 = 127\,775 \Omega$
- $I_B = (3.127 - 0.7) / 127\,775 = 2.427 / 127\,775 = \mathbf{19.0 \mu A}$
- $I_C = 120 \times 19.0 \mu A = \mathbf{2.28 \text{ mA}}$
- $I_E = 121 \times 19.0 \mu A = 2.298 \text{ mA}$

$$V_E = (2.298 \text{ mA})(1 \text{ k}\Omega) = \mathbf{2.30 \text{ V}}$$

$$7. V_{CE} = 18 - (2.28 \text{ mA})(3.3 \text{ k}\Omega) - (2.298 \text{ mA})(1 \text{ k}\Omega) = 18 - 7.52 - 2.30 = \mathbf{8.18 \text{ V}}$$

$$8. V_B = 2.30 + 0.7 = \mathbf{3.00 \text{ V}}$$

Comparison:

Quantity	Approximate (Q21)	Exact (Q22)	Difference
I_B	20.23 μA	19.00 μA	6.5 %
I_C	2.43 mA	2.28 mA	6.5 %
V_{CE}	7.56 V	8.18 V	7.5 %
V_E	2.43 V	2.30 V	5.7 %
V_B	3.13 V	3.00 V	4.3 %

Answer:

- Exact: $I_C = 2.28 \text{ mA}$, $V_{CE} = 8.18 \text{ V}$, $I_B = 19.0 \mu\text{A}$, $V_E = 2.30 \text{ V}$, $V_B = 3.00 \text{ V}$
- Yes, the approximate approach is valid.** Every quantity differs by less than about 8 %, which is within normal engineering tolerance, and β itself varies far more than that between transistors. Since Eq. (4.33) was satisfied, the approximation is justified.

Collector-Feedback Configuration

Question 27 — Fig. 4.129

Problem Type: Collector-Feedback Bias (Type 4)

Circuit values read from Fig. 4.129: $V_{CC} = 16 \text{ V}$, $R_C = 3.6 \text{ k}\Omega$, $R_F = 270 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$, $\beta = 120$

Given:

- $V_{CC} = 16 \text{ V}$, $R_C = 3.6 \text{ k}\Omega$, $R_F = 270 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$, $\beta = 120$

Find:

- I_B , I_C , V_C

Formula:

$$I_B = (V_{CC} - 0.7) / [R_F + \beta(R_C + R_E)] ; \quad I_C = \beta I_B ; \quad V_C = V_{CC} - I_C R_C$$

Solution:

- $R_C + R_E = 3600 + 1200 = 4800 \Omega$
 $\beta(R_C + R_E) = 120 \times 4800 = 576\,000 \Omega$
- Denominator = $270\,000 + 576\,000 = 846\,000 \Omega$
- $I_B = (16 - 0.7) / 846\,000 = 15.3 / 846\,000 = \mathbf{18.09 \mu\text{A}}$
- $I_C = 120 \times 18.09 \mu\text{A} = \mathbf{2.17 \text{ mA}}$
- $V_C = 16 - (2.17 \text{ mA})(3.6 \text{ k}\Omega) = 16 - 7.81 = \mathbf{8.19 \text{ V}}$

Answer:

- (a) $I_B = 18.09 \mu\text{A}$ (b) $I_C = 2.17 \text{ mA}$ (c) $V_C = 8.19 \text{ V}$

Question 29 — Fig. 4.130

Problem Type: Collector-Feedback (voltage-feedback) Bias (Type 4)

Circuit values read from Fig. 4.130: $V_{CC} = 30 \text{ V}$, $R_C = 8.2 \text{ k}\Omega$, $R_E = 1.8 \text{ k}\Omega$, feedback path = $330 \text{ k}\Omega + 220 \text{ k}\Omega$ in series, $\beta = 180$

Given:

- $V_{CC} = 30 \text{ V}$, $R_C = 8.2 \text{ k}\Omega$, $R_E = 1.8 \text{ k}\Omega$, $\beta = 180$
- The feedback path has $330 \text{ k}\Omega$ and $220 \text{ k}\Omega$ in series, with a $5 \text{ }\mu\text{F}$ capacitor to ground at their junction

Find:

- I_C , V_C , V_E , V_{CE}

Formula:

For DC, capacitors are OPEN $\rightarrow R_F = 330 \text{ k}\Omega + 220 \text{ k}\Omega = 550 \text{ k}\Omega$
 $I_B = (V_{CC} - 0.7) / [R_F + \beta(R_C + R_E)]$

Solution:

1. All capacitors ($10 \text{ }\mu\text{F}$, $5 \text{ }\mu\text{F}$) are open circuits for DC.
The $5 \text{ }\mu\text{F}$ to ground carries no DC current, so the two feedback resistors are simply in series:
 $R_F = 330 + 220 = \mathbf{550 \text{ k}\Omega}$
2. $R_C + R_E = 8200 + 1800 = 10\,000 \text{ }\Omega$
 $\beta(R_C + R_E) = 180 \times 10\,000 = 1\,800\,000 \text{ }\Omega$
3. Denominator = $550\,000 + 1\,800\,000 = 2\,350\,000 \text{ }\Omega$
4. $I_B = (30 - 0.7) / 2\,350\,000 = 29.3 / 2\,350\,000 = 12.47 \text{ }\mu\text{A}$
5. $I_C = 180 \times 12.47 \text{ }\mu\text{A} = \mathbf{2.24 \text{ mA}}$
6. $V_C = 30 - (2.244 \text{ mA})(8.2 \text{ k}\Omega) = 30 - 18.40 = \mathbf{11.60 \text{ V}}$
7. $V_E = (2.244 \text{ mA})(1.8 \text{ k}\Omega) = \mathbf{4.04 \text{ V}}$
8. $V_{CE} = 11.60 - 4.04 = \mathbf{7.56 \text{ V}}$

Answer:

- (a) $I_C = 2.24 \text{ mA}$ (b) $V_C = 11.60 \text{ V}$ (c) $V_E = 4.04 \text{ V}$ (d) $V_{CE} = 7.56 \text{ V}$

Question 33 — Fig. 4.133

Problem Type: Collector-Feedback $\rightarrow$ Reverse problem (Type 8)

Circuit values read from Fig. 4.133: $V_{CC} = 18 \text{ V}$, $R_C = 2.2 \text{ k}\Omega$, $R_F = 330 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$; β unknown, $V_B = 4 \text{ V}$ given in the question

Given:

- $V_{CC} = 18 \text{ V}$, $R_C = 2.2 \text{ k}\Omega$, $R_F = 330 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$, $V_B = 4 \text{ V}$

Find:

- V_E , I_C , V_C , V_{CE} , I_B , β

Formula:

$V_E = V_B - 0.7$; $I_C \approx I_E = V_E / R_E$; $V_C = V_{CC} - I_C R_C$
 $I_B = (V_C - V_B) / R_F$; $\beta = I_C / I_B$

Solution:

1. $V_E = 4 - 0.7 = \mathbf{3.3 \text{ V}}$
2. $I_E = 3.3 / 1.2 \text{ k}\Omega = 2.75 \text{ mA}$, so $I_C \approx \mathbf{2.75 \text{ mA}}$
3. $V_C = 18 - (2.75 \text{ mA})(2.2 \text{ k}\Omega) = 18 - 6.05 = \mathbf{11.95 \text{ V}}$
4. $V_{CE} = V_C - V_E = 11.95 - 3.3 = \mathbf{8.65 \text{ V}}$
5. R_F runs from the collector to the base, so the voltage across it is $V_C - V_B$:
 $I_B = (11.95 - 4) / 330 \text{ k}\Omega = 7.95 / 330\,000 = \mathbf{24.09 \text{ }\mu\text{A}}$
6. $\beta = I_C / I_B = 2.75 \text{ mA} / 24.09 \text{ }\mu\text{A} = \mathbf{114.2}$

Answer:

- (a) $V_E = 3.3 \text{ V}$ (b) $I_C = 2.75 \text{ mA}$ (c) $V_C = 11.95 \text{ V}$

- (d) $V_{CE} = 8.65 \text{ V}$ (e) $I_B = 24.09 \mu\text{A}$ (f) $\beta = 114.2$

Emitter-Follower Configuration

Question 34 — Fig. 4.134

Problem Type: Emitter Follower with split supply (Type 5)

Circuit values read from Fig. 4.134: +6 V top rail, $R_B = 330 \text{ k}\Omega$ to base, collector grounded, $R_E = 1.2 \text{ k}\Omega$ to a -6 V rail, $\beta = 120$

Given:

- Top supply +6 V, $R_B = 330 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$, bottom supply -6 V, $\beta = 120$

Find:

- V_E and I_E

Formula:

KVL base-emitter loop:
 $V_{\text{top}} + |V_{\text{bottom}}| - 0.7 = I_B[R_B + (\beta+1)R_E]$
 $V_E = -|V_{\text{bottom}}| + I_E R_E$

Solution:

1. Total driving voltage around the base loop = $6 + 6 - 0.7 = \mathbf{11.3 \text{ V}}$
(both supplies add, because the loop goes from +6 V down to -6 V)
2. Denominator = $330\,000 + 121(1200) = 330\,000 + 145\,200 = 475\,200 \Omega$
3. $I_B = 11.3 / 475\,200 = 23.78 \mu\text{A}$
4. $I_E = 121 \times 23.78 \mu\text{A} = \mathbf{2.877 \text{ mA}}$
5. V_E is measured from ground. Starting at the -6 V rail and rising through R_E :
 $V_E = -6 + (2.877 \text{ mA})(1.2 \text{ k}\Omega) = -6 + 3.45 = \mathbf{-2.55 \text{ V}}$

Answer:

- $I_E = 2.877 \text{ mA}$, $V_E = -2.55 \text{ V}$

Watch out: V_E is negative here. The emitter sits below ground because the bottom rail is at -6 V.

Question 35 — Fig. 4.135

Problem Type: Emitter Follower with voltage divider (Type 5)

Circuit values read from Fig. 4.135: $V_{CC} = 12 \text{ V}$, $R_1 = 22 \text{ k}\Omega$ (top), $R_2 = 82 \text{ k}\Omega$ (bottom), $R_E = 1.2 \text{ k}\Omega$, collector tied straight to V_{CC} , $\beta = 110$

Given:

- $V_{CC} = 12 \text{ V}$, $R_1 = 22 \text{ k}\Omega$, $R_2 = 82 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$, $\beta = 110$
- No R_C — the collector connects directly to V_{CC}

Find:

- (a) I_B , I_C , I_E (b) V_B , V_C , V_E (c) V_{BC} , V_{CE}

Formula:

$R_{Th} = R_1 R_2 / (R_1 + R_2)$; $E_{Th} = V_{CC} R_2 / (R_1 + R_2)$
 $I_B = (E_{Th} - 0.7) / [R_{Th} + (\beta+1)R_E]$

Solution — part (a):

1. $R_{Th} = (22 \times 82) / (22 + 82) = 1804 / 104 = \mathbf{17.35 \text{ k}\Omega}$
2. $E_{Th} = 12 \times 82 / 104 = 984 / 104 = \mathbf{9.46 \text{ V}}$

3. Denominator = $17\,346 + 111(1200) = 17\,346 + 133\,200 = 150\,546\ \Omega$
4. $I_B = (9.462 - 0.7) / 150\,546 = 8.762 / 150\,546 = \mathbf{58.2\ \mu A}$
5. $I_C = 110 \times 58.2\ \mu A = \mathbf{6.40\ mA}$
6. $I_E = 111 \times 58.2\ \mu A = \mathbf{6.46\ mA}$

Solution — part (b):

7. $V_E = I_E R_E = (6.46\ mA)(1.2\ k\Omega) = \mathbf{7.75\ V}$
8. $V_B = V_E + 0.7 = \mathbf{8.45\ V}$
9. $V_C = V_{CC} = \mathbf{12\ V}$ (no R_C , so no drop)

Solution — part (c):

10. $V_{BC} = V_B - V_C = 8.45 - 12 = \mathbf{-3.55\ V}$
11. $V_{CE} = V_C - V_E = 12 - 7.75 = \mathbf{4.25\ V}$

Answer:

- (a) $I_B = 58.2\ \mu A$, $I_C = 6.40\ mA$, $I_E = 6.46\ mA$
- (b) $V_B = 8.45\ V$, $V_C = 12\ V$, $V_E = 7.75\ V$
- (c) $V_{BC} = -3.55\ V$, $V_{CE} = 4.25\ V$

V_{BC} is negative, which is exactly right — the base-collector junction must be reverse-biased in the active region.

Common-Base Configuration

Question 36 — Fig. 4.136

Problem Type: Common Base with split supply (Type 6)

Circuit values read from Fig. 4.136: +16 V through $R_C = 12\ k\Omega$ to the collector, base to ground through $9.1\ k\Omega$, emitter through $R_E = 15\ k\Omega$ to -12 V, $\beta = 80$

Given:

- +16 V supply, $R_C = 12\ k\Omega$, $R_B = 9.1\ k\Omega$ (base to ground), $R_E = 15\ k\Omega$, -12 V supply, $\beta = 80$

Find:

- I_B , I_C , V_{CE} , V_C

Formula:

KVL base-emitter loop (from ground, through R_B , the junction, R_E , to -12 V):
 $12 - 0.7 = I_B[R_B + (\beta+1)R_E]$

Solution:

1. Driving voltage in the base loop = $12 - 0.7 = \mathbf{11.3\ V}$
2. Denominator = $9100 + 81(15\,000) = 9100 + 1\,215\,000 = 1\,224\,100\ \Omega$
3. $I_B = 11.3 / 1\,224\,100 = \mathbf{9.23\ \mu A}$
4. $I_C = 80 \times 9.23\ \mu A = \mathbf{0.739\ mA}$
5. $I_E = 81 \times 9.23\ \mu A = 0.748\ mA$
6. $V_C = 16 - (0.7385\ mA)(12\ k\Omega) = 16 - 8.86 = \mathbf{7.14\ V}$
7. $V_E = -12 + (0.7477\ mA)(15\ k\Omega) = -12 + 11.22 = -0.78\ V$
8. $V_{CE} = V_C - V_E = 7.14 - (-0.78) = \mathbf{7.92\ V}$

Answer:

- (a) $I_B = 9.23\ \mu A$ (b) $I_C = 0.739\ mA$ (c) $V_{CE} = 7.92\ V$ (d) $V_C = 7.14\ V$

Question 37 — Fig. 4.137

Problem Type: Common Base, base grounded (Type 6)

Circuit values read from Fig. 4.137: emitter through $2.2\text{ k}\Omega$ to -8 V , base grounded, collector through $1.8\text{ k}\Omega$ to $+10\text{ V}$. β is not given.

Given:

- $R_E = 2.2\text{ k}\Omega$ to -8 V , base grounded, $R_C = 1.8\text{ k}\Omega$ to $+10\text{ V}$

Find:

- I_E , V_C , V_{CE}

Formula:

$$V_E = V_B - 0.7 = -0.7\text{ V} \quad (\text{base grounded})$$

$$I_E = (V_E - V_{EE})/R_E ; \quad I_C \approx I_E ; \quad V_C = V_{CC} - I_C R_C$$

Solution:

- Base is grounded, so $V_B = 0$ and $V_E = 0 - 0.7 = -0.7\text{ V}$
- Voltage across $R_E = V_E - (-8) = -0.7 + 8 = 7.3\text{ V}$
 $I_E = 7.3 / 2.2\text{ k}\Omega = 3.32\text{ mA}$
- β is not given, so take $I_C \approx I_E = 3.32\text{ mA}$
 $V_C = 10 - (3.318\text{ mA})(1.8\text{ k}\Omega) = 10 - 5.97 = 4.03\text{ V}$
- $V_{CE} = V_C - V_E = 4.03 - (-0.7) = 4.73\text{ V}$

Answer:

- (a) $I_E = 3.32\text{ mA}$ (b) $V_C = 4.03\text{ V}$ (c) $V_{CE} = 4.73\text{ V}$

Question 38 — Fig. 4.138

Problem Type: Common Base $\rightarrow$ Reverse problem (Types 6 + 8)

Circuit values read from Fig. 4.138: 14 V supply through R_C (unknown) to the collector, $V_C = 8\text{ V}$, base held at 4 V , $R_E = 1.1\text{ k}\Omega$ to ground, $\beta = 90$

Given:

- $V_{CC} = 14\text{ V}$, $V_C = 8\text{ V}$, $V_B = 4\text{ V}$, $R_E = 1.1\text{ k}\Omega$, $\beta = 90$

Find:

- (a) R_C (b) I_B and I_E (c) V_{BC} and V_{CE}

Formula:

$$V_E = V_B - 0.7 ; \quad I_E = V_E/R_E ; \quad I_B = I_E/(\beta+1) ; \quad I_C = \beta I_B$$

$$R_C = (V_{CC} - V_C)/I_C$$

Solution — part (a):

- $V_E = 4 - 0.7 = 3.3\text{ V}$
- $I_E = 3.3 / 1.1\text{ k}\Omega = 3\text{ mA}$
- $I_B = 3\text{ mA} / 91 = 32.97\text{ }\mu\text{A}$
 $I_C = 90 \times 32.97\text{ }\mu\text{A} = 2.967\text{ mA}$
- Drop across $R_C = 14 - 8 = 6\text{ V}$
 $R_C = 6 / 2.967\text{ mA} = 2.02\text{ k}\Omega$

Solution — part (b):

- $I_B = 32.97\text{ }\mu\text{A}$, $I_E = 3\text{ mA}$

Solution — part (c):

- $V_{BC} = V_B - V_C = 4 - 8 = -4\text{ V}$
- $V_{CE} = V_C - V_E = 8 - 3.3 = 4.7\text{ V}$

Answer:

- (a) $R_C = 2.02 \text{ k}\Omega$ (b) $I_B = 32.97 \text{ }\mu\text{A}$, $I_E = 3 \text{ mA}$ (c) $V_{BC} = -4 \text{ V}$, $V_{CE} = 4.7 \text{ V}$

Design Operations

Question 41 — Fixed-bias design

Problem Type: Design $\rightarrow$ Fixed bias (Type 9)

Given:

- $V_{CC} = 12 \text{ V}$, $\beta = 80$, $I_{CQ} = 2.5 \text{ mA}$, $V_{CEQ} = 6 \text{ V}$

Find:

- R_C and R_B , rounded to standard values

Formula:

$$R_C = (V_{CC} - V_{CEQ}) / I_{CQ} ; \quad I_B = I_{CQ} / \beta ; \quad R_B = (V_{CC} - 0.7) / I_B$$

Solution:

1. Voltage across $R_C = 12 - 6 = 6 \text{ V}$
 $R_C = 6 / 2.5 \text{ mA} = 2.4 \text{ k}\Omega \rightarrow$ standard value **2.4 k Ω** (exact match)
2. $I_B = 2.5 \text{ mA} / 80 = 31.25 \text{ }\mu\text{A}$
3. $R_B = (12 - 0.7) / 31.25 \text{ }\mu\text{A} = 11.3 / 31.25 \times 10^{-6} = 361.6 \text{ k}\Omega$
 Nearest standard value: **360 k Ω**

Answer:

- $R_C = 2.4 \text{ k}\Omega$ (standard), $R_B = 361.6 \text{ k}\Omega \rightarrow$ use **360 k Ω** (standard)

Question 42 — Emitter-stabilized design

Problem Type: Design $\rightarrow$ Emitter bias (Type 9)

Given:

- $V_{CC} = 20 \text{ V}$, $I_{Csat} = 10 \text{ mA}$, $\beta = 120$, $R_C = 4R_E$
- Target: $I_{CQ} = \frac{1}{2}I_{Csat}$ and $V_{CEQ} = \frac{1}{2}V_{CC}$

Find:

- R_C , R_E , R_B

Formula:

$$I_{Csat} = V_{CC} / (R_C + R_E) \rightarrow R_C + R_E = V_{CC} / I_{Csat}$$

$$R_B = (V_{CC} - 0.7 - V_E) / I_B$$

Solution:

1. Targets: $I_{CQ} = \frac{1}{2}(10 \text{ mA}) = \mathbf{5 \text{ mA}}$, $V_{CEQ} = \frac{1}{2}(20) = \mathbf{10 \text{ V}}$
2. From saturation: $R_C + R_E = 20 / 10 \text{ mA} = 2000 \text{ }\Omega$
3. With $R_C = 4R_E$: $4R_E + R_E = 5R_E = 2000$
 $R_E = \mathbf{400 \text{ }\Omega} \rightarrow$ nearest standard **390 Ω**
 $R_C = 4(400) = \mathbf{1.6 \text{ k}\Omega} \rightarrow$ standard **1.6 k Ω** (exact match)
4. Check: $V_{CEQ} = 20 - (5 \text{ mA})(2 \text{ k}\Omega) = 20 - 10 = 10 \text{ V} \checkmark$
5. $I_B = 5 \text{ mA} / 120 = 41.67 \text{ }\mu\text{A}$
 $I_E = 121 \times 41.67 \text{ }\mu\text{A} = 5.042 \text{ mA}$
 $V_E = (5.042 \text{ mA})(400 \text{ }\Omega) = 2.02 \text{ V}$
6. $R_B = (20 - 0.7 - 2.02) / 41.67 \text{ }\mu\text{A} = 17.28 / 41.67 \times 10^{-6} = 414.8 \text{ k}\Omega$
 Nearest standard: **430 k Ω**

Answer:

- $R_E = 400 \, \Omega \rightarrow \mathbf{390 \, \Omega}$ (standard)
- $R_C = \mathbf{1.6 \, k\Omega}$ (standard)
- $R_B = 414.8 \, k\Omega \rightarrow \mathbf{430 \, k\Omega}$ (standard)

Question 43 — Voltage-divider design**Problem Type:** Design $\rightarrow$ Voltage-divider bias (Type 9)**Given:**

- $V_{CC} = 24 \, V$, $\beta = 110$, $I_{CQ} = 4 \, mA$, $V_{CEQ} = 8 \, V$, $V_E = \frac{1}{8}V_{CC}$

Find:

- R_E , R_C , R_1 , R_2

Formula:

$$V_E = V_{CC}/8 ; \quad R_E = V_E/I_C ; \quad R_C = (V_{CC} - V_{CEQ} - V_E)/I_C$$

$$R_2 \leq \beta R_E/10 ; \quad R_1 = V_{CC} R_2/V_B - R_2$$

Solution:

1. $V_E = 24/8 = \mathbf{3 \, V}$
2. $R_E = 3 / 4 \, mA = \mathbf{750 \, \Omega} \rightarrow$ standard $\mathbf{750 \, \Omega}$ (exact match)
3. Voltage across $R_C = 24 - 8 - 3 = 13 \, V$
 $R_C = 13 / 4 \, mA = 3.25 \, k\Omega \rightarrow$ nearest standard $\mathbf{3.3 \, k\Omega}$
4. $V_B = V_E + 0.7 = \mathbf{3.7 \, V}$
5. Choose R_2 using Eq. (4.33): $R_2 \leq \beta R_E/10 = (110 \times 750)/10 = 8250 \, \Omega$
 Pick standard $\mathbf{R_2 = 8.2 \, k\Omega}$ (just under the limit)
6. $R_1 + R_2 = V_{CC}R_2/V_B = (24 \times 8200)/3.7 = 196 \, 800/3.7 = 53.19 \, k\Omega$
 $R_1 = 53.19 - 8.2 = 44.99 \, k\Omega \rightarrow$ nearest standard $\mathbf{43 \, k\Omega}$
 (47 k Ω is almost equally close; either is defensible — state which you chose.)

Answer:

- $R_E = \mathbf{750 \, \Omega}$, $R_C = 3.25 \, k\Omega \rightarrow \mathbf{3.3 \, k\Omega}$, $R_2 = \mathbf{8.2 \, k\Omega}$, $R_1 = 44.99 \, k\Omega \rightarrow \mathbf{43 \, k\Omega}$

Current Source Circuits**Question 51 — Fig. 4.147****Problem Type:** Current source, fixed base supply (Type 10)

Circuit values read from Fig. 4.147: 28 V through a 2.2 k Ω load to the collector, base fed from +6 V through $R_B = 100 \, k\Omega$, $R_E = 1.2 \, k\Omega$ to ground, $\beta = 120$

Given:

- Load = 2.2 k Ω (from 28 V to the collector), base supply +6 V, $R_B = 100 \, k\Omega$, $R_E = 1.2 \, k\Omega$, $\beta = 120$

Find:

- The current through the 2.2 k Ω load

Formula:

$$I_B = (6 - 0.7)/[R_B + (\beta+1)R_E] ; \quad I = I_C = \beta I_B$$

Solution:

1. Denominator = $100 \, 000 + 121(1200) = 100 \, 000 + 145 \, 200 = 245 \, 200 \, \Omega$
2. $I_B = (6 - 0.7) / 245 \, 200 = 5.3 / 245 \, 200 = 21.62 \, \mu A$

3. $I = I_C = 120 \times 21.62 \mu\text{A} = \mathbf{2.59 \text{ mA}}$
4. Check the transistor is active: $V_C = 28 - (2.594 \text{ mA})(2.2 \text{ k}\Omega) = 28 - 5.71 = 22.29 \text{ V}$,
 $V_E = (2.615 \text{ mA})(1.2 \text{ k}\Omega) = 3.14 \text{ V} \rightarrow V_{CE} = 19.15 \text{ V}$, comfortably active ✓

Answer:

- Current through the $2.2 \text{ k}\Omega$ load = **2.59 mA**

The load value never entered the calculation — that is the defining feature of a current source.

Question 52 — Fig. 4.148

Problem Type: Current source with a resistor divider (Type 10)

Circuit values read from Fig. 4.148: base divider of $4.3 \text{ k}\Omega$ (to ground) and $4.3 \text{ k}\Omega$ (to -18 V), $R_E = 1.8 \text{ k}\Omega$ to -18 V , $\beta = 100$

Given:

- Base divider: $4.3 \text{ k}\Omega$ from base to ground, $4.3 \text{ k}\Omega$ from base to -18 V
- $R_E = 1.8 \text{ k}\Omega$ to -18 V , $\beta = 100$

Find:

- The current I

Formula:

$$R_{Th} = 4.3\text{k} \parallel 4.3\text{k} ; \quad E_{Th} = \text{divider between } 0 \text{ V and } -18 \text{ V}$$

$$I_B = (E_{Th} - V_{EE} - 0.7) / [R_{Th} + (\beta + 1)R_E]$$

Solution:

1. The divider sits between 0 V and -18 V with two equal resistors, so the base sits halfway:
 $E_{Th} = -18 \times 4.3 / (4.3 + 4.3) = \mathbf{-9 \text{ V}}$
2. $R_{Th} = (4.3 \times 4.3) / (4.3 + 4.3) = \mathbf{2.15 \text{ k}\Omega}$
3. Base loop driving voltage = $E_{Th} - V_{EE} - 0.7 = -9 - (-18) - 0.7 = \mathbf{8.3 \text{ V}}$
4. Denominator = $2150 + 101(1800) = 2150 + 181\,800 = 183\,950 \Omega$
5. $I_B = 8.3 / 183\,950 = 45.12 \mu\text{A}$
6. $I = I_C = 100 \times 45.12 \mu\text{A} = \mathbf{4.51 \text{ mA}}$

Answer:

- $I = 4.51 \text{ mA}$

Using the quick approximation (ignoring I_B): $V_E = -9.7 \text{ V}$, $I_E = 8.3 / 1.8 \text{ k}\Omega = 4.61 \text{ mA}$. That is about 2 % off — the exact answer 4.51 mA is preferred.

Question 53 — Fig. 4.149

Problem Type: Zener-controlled current source (Type 10)

Circuit values read from Fig. 4.149: base held by a 5.1 V Zener referenced to -12 V , $1.5 \text{ k}\Omega$ from the base node to ground, $R_E = 1.2 \text{ k}\Omega$ to -12 V , $\beta = 200$

Given:

- $V_Z = 5.1 \text{ V}$ (between the base and the -12 V rail), $R_E = 1.2 \text{ k}\Omega$, $V_{EE} = -12 \text{ V}$, $\beta = 200$

Find:

- The current I

Formula:

$$V_B = V_{EE} + V_Z \quad (\text{Zener fixes the base voltage directly})$$

$$V_E = V_B - 0.7 ; \quad I_E = (V_E - V_{EE})/R_E ; \quad I = I_C = \beta/(\beta+1) \times I_E$$

Solution:

1. The Zener pins the base 5.1 V above the -12 V rail:
 $V_B = -12 + 5.1 = \mathbf{-6.9 \text{ V}}$
2. $V_E = -6.9 - 0.7 = \mathbf{-7.6 \text{ V}}$
3. Voltage across $R_E = V_E - (-12) = -7.6 + 12 = 4.4 \text{ V}$
 $I_E = 4.4 / 1.2 \text{ k}\Omega = \mathbf{3.67 \text{ mA}}$
4. $I = I_C = \alpha I_E = (200/201)(3.667 \text{ mA}) = \mathbf{3.65 \text{ mA}}$

Answer:

- $I = 3.65 \text{ mA}$

The $1.5 \text{ k}\Omega$ resistor only supplies the Zener's bias current — it does not appear in the load-current calculation.

Chapter 1 — Semiconductor Diodes

Question 15

Problem Type: (a) Thermal voltage (Type 1); (b) Diode current from Shockley's equation (Type 2)

Given:

- (a) $T = 20\text{ }^{\circ}\text{C}$
- (b) $I_s = 40\text{ nA}$, $n = 2$, $V_D = 0.5\text{ V}$, same temperature

Find:

- (a) V_T (b) I_D

Formula:

$$T_K = T_C + 273 ; \quad V_T = (8.62 \times 10^{-5}) T_K$$

$$I_D = I_s (e^{(V_D/(nV_T))} - 1)$$

Solution — part (a):

1. $T_K = 20 + 273 = 293\text{ K}$
2. $V_T = (8.62 \times 10^{-5})(293) = 0.02525\text{ V} \approx \mathbf{25.27\text{ mV}}$

Solution — part (b):

3. $nV_T = 2 \times 0.02527 = 0.05054\text{ V}$
4. Exponent $= V_D/(nV_T) = 0.5 / 0.05054 = 9.893$
5. $e^{9.893} = 19\,785$
6. $I_D = (40 \times 10^{-9})(19\,785 - 1) = (40 \times 10^{-9})(19\,784) = 7.91 \times 10^{-4}\text{ A}$

Answer:

- (a) $V_T = 25.27\text{ mV}$
- (b) $I_D = 0.791\text{ mA}$

Question 16

Problem Type: Thermal voltage + diode current at a new temperature (Types 1 + 2)

Value flagged: your document states " I_s has increased to **5.0 mA**". A reverse saturation current of 5 mA is extremely large for a diode (typical values are nA to μA), so this looks like a " μ " that was lost when the text was copied. **I have not assumed either way — both answers are below.** Use whichever matches your question paper.

Given:

- $T = 100\text{ }^{\circ}\text{C}$, $n = 2$, $V_D = 0.5\text{ V}$, $I_s = 5.0\text{ mA}$ (as written) or $5.0\text{ }\mu\text{A}$ (if the μ was lost)

Find:

- V_T and I_D

Formula:

$$V_T = (8.62 \times 10^{-5}) T_K ; \quad I_D = I_s (e^{(V_D/(nV_T))} - 1)$$

Solution:

1. $T_K = 100 + 273 = 373\text{ K}$
2. $V_T = (8.62 \times 10^{-5})(373) = 0.03215\text{ V} \approx \mathbf{32.17\text{ mV}}$
3. $nV_T = 2 \times 0.03217 = 0.06434\text{ V}$
4. Exponent $= 0.5 / 0.06434 = 7.771$
5. $e^{7.771} = 2370.6$

6. If $I_s = 5.0 \text{ mA}$ (as written):

$$I_D = (5 \times 10^{-3})(2369.6) = \mathbf{11.85 \text{ A}}$$

7. If $I_s = 5.0 \text{ }\mu\text{A}$:

$$I_D = (5 \times 10^{-6})(2369.6) = \mathbf{11.85 \text{ mA}}$$

Answer:

- $V_T = 32.17 \text{ mV}$
- $I_D = \mathbf{11.85 \text{ A}}$ using the value as written (5.0 mA), or **11.85 mA** if I_s is 5.0 μA
- The method and every intermediate number are identical; only the final multiplier changes.

Question 17**Problem Type:** Reverse-bias diode current (Type 3)

Value flagged: your document states " $I_s \mathbf{0.1 \text{ mA}}$ ". As in Q16, this is likely 0.1 μA . Fortunately the *method* and the *form* of the answer are unaffected — the answer is simply $-I_s$, whatever its value.

Given:

- $T = 20 \text{ }^\circ\text{C}$, silicon diode, $n = 2$, $I_s = 0.1 \text{ mA}$ (as written), $V_D = -10 \text{ V}$

Find:

- **(a)** I_D **(b)** Is the result expected? Why?

Formula:

$$I_D = I_s (e^{(V_D/(nV_T))} - 1)$$

Solution — part (a):

1. V_T at $20 \text{ }^\circ\text{C} = 25.27 \text{ mV}$ (from Q15), so $nV_T = 0.05054 \text{ V}$
2. Exponent $= -10 / 0.05054 = \mathbf{-197.9}$
3. $e^{-197.9} \approx 1.2 \times 10^{-86}$ — effectively **zero**
4. $I_D = I_s(0 - 1) = \mathbf{-I_s = -0.1 \text{ mA}}$ (or $-0.1 \text{ }\mu\text{A}$ if I_s is in μA)

Solution — part (b):

5. **Yes, this is exactly what we expect.** Under reverse bias the exponential term collapses to zero, so the equation reduces to $I_D = -I_s$. The diode blocks, and only the tiny reverse saturation (leakage) current flows, in the reverse direction. The minus sign just shows the direction is opposite to normal forward conduction.
6. Note also that the answer does **not** depend on how large the reverse voltage is — anything past about -0.5 V gives the same result. That is why the reverse characteristic is drawn as a flat horizontal line.

Answer:

- **(a)** $I_D = -I_s = -0.1 \text{ mA}$ (as written)
- **(b)** Yes — expected. The exponential term vanishes under reverse bias, leaving only the reverse saturation current.

Question 18**Problem Type:** Find I_s from a known I_D (Type 4)**Given:**

- $I_D = 8 \text{ mA}$, $n = 1$, $V_D = 0.5 \text{ V}$, $T = 25 \text{ }^\circ\text{C}$ (room temperature)

Find:

- I_s

Formula:

$$I_s = I_D / (e^{(V_D/(nV_T))} - 1)$$

Solution:

1. $T_K = 25 + 273 = 298 \text{ K}$
2. $V_T = (8.62 \times 10^{-5})(298) = 0.02569 \text{ V} \approx 25.70 \text{ mV}$
3. $n = 1$, so $nV_T = 0.02570 \text{ V}$
4. $\text{Exponent} = 0.5 / 0.02570 = 19.453$
5. $e^{19.453} = 2.809 \times 10^8$
6. $I_s = (8 \times 10^{-3}) / (2.809 \times 10^8 - 1) = 2.85 \times 10^{-11} \text{ A}$

Answer:

- $I_s = 2.85 \times 10^{-11} \text{ A} = \mathbf{28.5 \text{ pA}}$

A picoamp-scale answer is exactly what you should expect. If you get milliamps, you divided the wrong way round.

Question 19

Problem Type: Find V_D from a known I_D (Type 5)

Given:

- $I_D = 6 \text{ mA}$, $V_T = 26 \text{ mV}$ (given directly), $n = 1$, $I_s = 1 \text{ nA}$

Find:

- V_D

Formula:

$$V_D = n V_T \ln(I_D / I_s + 1)$$

Solution:

1. $I_D / I_s = (6 \times 10^{-3}) / (1 \times 10^{-9}) = 6 \times 10^6$
2. Add 1: $6\,000\,001 \approx 6 \times 10^6$ (the +1 is negligible here)
3. $\ln(6 \times 10^6) = \mathbf{15.607}$
4. $V_D = (1)(0.026)(15.607) = 0.4058 \text{ V}$

Answer:

- $V_D = \mathbf{0.406 \text{ V}}$

Sanity check: 0.406 V is a believable forward voltage for a diode. Use **ln**, not $\log_{10}$ — $\log_{10}$ would give 0.176 V, which is wrong.

Chapter 2 — Diode Applications

Series Diode Configurations

Question 5 — Fig. 2.155

Problem Type: Series diode circuits → ON/OFF test (Types 1, 2)

Note on Fig. 2.155(a): the first digit of the battery label is cut off at the left edge of the image in your document (it reads "...2 V"). **This does not change the answer** — the diode is reverse-biased regardless of the magnitude, so $I = 0$ A either way.

Circuit values read from Fig. 2.155:

- **(a)** Battery with – at the top, + at the bottom (grounded), Si diode pointing right, $10\ \Omega$
- **(b)** $10\ \Omega$ branch, Si diode pointing right, a second Si diode with its cathode at the top node in series with $20\ \Omega$ to ground, and a 20 V battery (– left, + right) whose + side is grounded
- **(c)** 10 V battery (+ top, – bottom grounded), two Si diodes in the middle branch pointing in **opposite** directions, $10\ \Omega$ on the right

Find:

- The current I in each configuration

Formula:

Check diode direction first. If reverse-biased → $I = 0$.
Otherwise: $I = (\text{driving voltage} - \Sigma V_K) / \Sigma R$

Solution — (a):

1. The battery has – on top and + at the bottom, and the bottom is grounded.
So the top node = –(battery value), i.e. it is **negative**.
2. The diode's anode faces that negative node; its cathode side reaches ground through the $10\ \Omega$.
3. Anode is more negative than the cathode → the diode is **reverse-biased**.
4. Therefore **$I = 0\text{ A}$** .

Solution — (b):

5. The 20 V battery has + grounded and – at the top node, so the top node A sits at **-20 V** .
6. Both Si diodes point **toward** node A (the horizontal one from the $10\ \Omega$ branch, the vertical one from the $20\ \Omega$ branch), so both are forward-biased: current flows from ground, up through the resistors, into node A, and back through the battery.
7. Left branch: the node above the $10\ \Omega$ sits at $V_A + 0.7 = -20 + 0.7 = -19.3\text{ V}$
 $I_1 = (0 - (-19.3)) / 10 = \mathbf{1.93\text{ A}}$
8. Middle branch: the node above the $20\ \Omega$ also sits at -19.3 V
 $I_2 = (0 - (-19.3)) / 20 = \mathbf{0.965\text{ A}}$
9. The marked I is the total returning through the battery:
 $I = I_1 + I_2 = 1.93 + 0.965 = \mathbf{2.90\text{ A}}$

Solution — (c):

10. The two Si diodes in the middle branch point in **opposite** directions (one conducts down, the other only up).
11. Opposing diodes in series → that branch is an **open circuit**, no current can flow through it.
12. What remains is simply the 10 V battery across the $10\ \Omega$:
 $I = 10 / 10 = \mathbf{1\text{ A}}$

Answer:

- **(a)** $I = 0\text{ A}$ (diode reverse-biased)
- **(b)** $I = 2.90\text{ A}$ (1.93 A through the $10\ \Omega$ + 0.965 A through the $20\ \Omega$)
- **(c)** $I = 1\text{ A}$ (middle branch is dead)

Question 6 — Fig. 2.156**Problem Type:** Series diode circuit → find V_o and I_D (Type 1)**Circuit values read from Fig. 2.156:**

- (a) -5 V source, Si diode with its **cathode facing the -5 V side**, output node, 2.2 k Ω to ground
- (b) $+8$ V, 1.2 k Ω , output node, 4.7 k Ω , Si diode pointing down, -6 V

Given / Find:

- V_o and I_D for each network

Formula:

$$I = (\Sigma V - \Sigma V_K) / \Sigma R ; \quad V_o \text{ from the node voltages}$$

Solution — (a):

1. The diode's cathode is at -5 V and its anode is at the output node, so current flows **from the output node toward the -5 V source** (right to left), matching the I_D arrow.
2. The diode is forward-biased. Across it: $V_{\text{anode}} - V_{\text{cathode}} = 0.7$
 $V_o - (-5) = 0.7 \rightarrow V_o = \mathbf{-4.3 \text{ V}}$
3. Current through the 2.2 k Ω (from ground down to V_o):
 $I_D = (0 - (-4.3)) / 2.2 \text{ k}\Omega = 4.3 / 2200 = \mathbf{1.96 \text{ mA}}$

Solution — (b):

4. Top is $+8$ V, bottom is -6 V, and the diode points downward → forward-biased.
5. Total driving voltage = $8 - (-6) = 14$ V; subtract the diode drop: $14 - 0.7 = 13.3$ V
6. Total resistance = $1.2 \text{ k}\Omega + 4.7 \text{ k}\Omega = 5.9 \text{ k}\Omega$
 $I_D = 13.3 / 5900 = \mathbf{2.25 \text{ mA}}$
7. $V_o = 8 - I_D(1.2 \text{ k}\Omega) = 8 - (2.254 \text{ mA})(1200) = 8 - 2.70 = \mathbf{5.30 \text{ V}}$

Answer:

- (a) $V_o = -4.3$ V, $I_D = 1.96$ mA
- (b) $V_o = 5.30$ V, $I_D = 2.25$ mA

Question 7 — Fig. 2.157**Problem Type:** Series diode circuit with mixed materials (Type 1)**Circuit values read from Fig. 2.157:**

- (a) 12 V, Si diode, Ge diode (both pointing right), 2 k Ω , output node, 10 k Ω to ground
- (b) $+10$ V, 1.2 k Ω , Si diode pointing right, output node, 4.7 k Ω , then a **$+10$ V** source at the bottom

Given / Find:

- V_o for each network

Formula:

$$I = (V - V_K(\text{Si}) - V_K(\text{Ge})) / \Sigma R ; \quad V_o = I \times R$$

Solution — (a):

1. Both diodes point the same way and the 12 V drives them forward → both conduct.
2. Total diode drop = 0.7 (Si) + 0.3 (Ge) = $\mathbf{1.0 \text{ V}}$
3. $I = (12 - 1.0) / (2 \text{ k}\Omega + 10 \text{ k}\Omega) = 11 / 12 \text{ 000} = \mathbf{0.917 \text{ mA}}$
4. $V_o = I \times 10 \text{ k}\Omega = (0.9167 \text{ mA})(10 \text{ 000}) = \mathbf{9.17 \text{ V}}$

Solution — (b):

5. **Test the diode first.** Assume it is off: no current flows, so there is no drop across either resistor. Then the left of the diode sits at 10 V and the right (output node) sits at 10 V — from the bottom source.
6. Voltage across the diode = $10 - 10 = 0$ V, which is less than 0.7 V → the diode is **OFF**.
7. Confirm by assuming it is on: KVL gives $10 - I(1.2 \text{ k}) - 0.7 - I(4.7 \text{ k}) - 10 = 0 \rightarrow I = -0.119 \text{ mA}$. A negative current contradicts the diode's direction, so it really is off.
8. With no current, there is no drop across the 4.7 kΩ, so $V_o =$ the bottom source = **10 V**

Answer:

- (a) $V_o = 9.17 \text{ V}$
- (b) $V_o = 10 \text{ V}$ (diode off — the two 10 V sources cancel, leaving nothing to forward-bias it)

Question 9 — Fig. 2.159

Problem Type: (a) Series diodes with two outputs; (b) opposing diodes (Types 1, 2)

Circuit values read from Fig. 2.159:

- (a) +12 V, Si diode pointing right, node V_{o1} , 4.7 kΩ, node V_{o2} , GaAs diode pointing down to ground
- (b) -10 V, Ge diode pointing right, Si diode with its **cathode on the left** (i.e. pointing left), node V_{o1} , 1.2 kΩ, node V_{o2} , 3.3 kΩ to ground

Given / Find:

- V_{o1} and V_{o2} for each network

Formula:

- (a) $I = (V - V_K(\text{Si}) - V_K(\text{GaAs}))/R$
 (b) Opposing diodes → open circuit → $I = 0$

Solution — (a):

1. Both diodes are forward-biased by the +12 V.
2. V_{o1} is right after the Si diode: $V_{o1} = 12 - 0.7 = 11.3 \text{ V}$
3. V_{o2} is across the GaAs diode to ground, so it equals that diode's drop:
 $V_{o2} = 1.2 \text{ V}$
4. Check with the current: $I = (12 - 0.7 - 1.2)/4.7 \text{ k}\Omega = 10.1/4700 = 2.15 \text{ mA}$
 and $V_{o1} - V_{o2} = 11.3 - 1.2 = 10.1 \text{ V}$ across the 4.7 kΩ ✓

Solution — (b):

5. The Ge diode points **right** (anode at the -10 V side); the Si diode points **left** (cathode on the left). They face in **opposite** directions.
6. Opposing diodes in series → **open circuit**, $I = 0$.
7. With no current, there is no drop across the 3.3 kΩ or the 1.2 kΩ, and both output nodes are tied to ground through those resistors:
 $V_{o2} = 0 \text{ V}$, and $V_{o1} = V_{o2} = 0 \text{ V}$

Answer:

- (a) $V_{o1} = 11.3 \text{ V}$, $V_{o2} = 1.2 \text{ V}$
- (b) $V_{o1} = 0 \text{ V}$, $V_{o2} = 0 \text{ V}$ (back-to-back diodes block in both directions)

Question 13 — Fig. 2.163

Problem Type: Parallel diode branches → KCL at the output node (Type 3)

Circuit values read from Fig. 2.163: +10 V feeding two parallel branches — an upper branch with a Si diode and 2 kΩ, a lower branch with a GaAs diode and 2 kΩ — joining at node V_o , which has a 2 kΩ to ground

Given:

- $V = 10 \text{ V}$, Si branch: 0.7 V + 2 kΩ; GaAs branch: 1.2 V + 2 kΩ; load 2 kΩ to ground

Find:

- V_o and I_D (the current in the Si branch, as marked)

Formula:

Assume both ON, then KCL at V_o :
 $(10 - 0.7 - V_o)/2k + (10 - 1.2 - V_o)/2k = V_o/2k$

Solution:

1. Assume both diodes conduct. Write KCL at the output node (current in = current out):
 $(10 - 0.7 - V_o)/2k + (10 - 1.2 - V_o)/2k = V_o/2k$
2. All resistors are 2 k Ω , so multiply through by 2k:
 $(9.3 - V_o) + (8.8 - V_o) = V_o$
3. $18.1 - 2V_o = V_o \rightarrow 18.1 = 3V_o$
4. $V_o = 18.1/3 = \mathbf{6.03\text{ V}}$
5. Si branch (this is the marked I_D):
 $I_D = (9.3 - 6.033)/2\text{ k}\Omega = 3.267/2000 = \mathbf{1.63\text{ mA}}$
6. **Check both assumptions:** GaAs branch = $(8.8 - 6.033)/2\text{ k}\Omega = 1.38\text{ mA}$. Both currents are positive, so both diodes really do conduct ✓
7. Check KCL: $1.63 + 1.38 = 3.02\text{ mA} = V_o/2k = 6.033/2000 = 3.02\text{ mA}$ ✓

Answer:

- $V_o = \mathbf{6.03\text{ V}}$, I_D (Si branch) = $\mathbf{1.63\text{ mA}}$
- (The GaAs branch carries 1.38 mA — both branches conduct because each has its own series resistor.)

Sinusoidal Inputs; Half-Wave Rectification**Question 22 — Fig. 2.168**

Problem Type: Half-wave rectifier, ideal diode $\rightarrow$ find V_m and sketch (Type 4)

Circuit values read from Fig. 2.168: $v_i \rightarrow$ ideal diode $\rightarrow$ output node $\rightarrow$ 2 k Ω to ground; the measured DC level is $V_{dc} = 2\text{ V}$

Given:

- Ideal diode, $R = 2\text{ k}\Omega$, $f = 60\text{ Hz}$, $V_{dc} = 2\text{ V}$

Find:

- The peak value of v_i , and sketches of v_i , v_d , i_d

Formula:

$V_{dc} = 0.318 V_m \rightarrow V_m = V_{dc} / 0.318$
 $T = 1/f$

Solution:

1. Work backwards from the DC level:
 $V_m = 2 / 0.318 = \mathbf{6.29\text{ V}}$
2. Period: $T = 1/60 = \mathbf{16.67\text{ ms}}$, so each half-cycle lasts 8.33 ms
3. Peak diode current: $i_{d(\text{peak})} = 6.29 / 2\text{ k}\Omega = \mathbf{3.14\text{ mA}}$

Sketches — what to draw:

Waveform	Positive half-cycle (diode ON)	Negative half-cycle (diode OFF)
v_i	sine rising to +6.29 V	sine falling to -6.29 V
v_d (across diode)	0 V (ideal diode = short)	follows v_i down to -6.29 V
i_d	half-sine peaking at 3.14 mA	0 mA

Answer:

- Peak value of v_i = **6.29 V**
- v_d = 0 during the positive half; $v_d = v_i$ (down to -6.29 V) during the negative half
- i_d = positive half-sine with a 3.14 mA peak, zero for the other half; $T = 16.67$ ms

The classic mistake: drawing v_d as zero everywhere. When the diode is OFF no current flows, so the **entire input appears across the diode**.

Question 23 — Fig. 2.168 with a silicon diode

Problem Type: Half-wave rectifier with $V_K = 0.7$ V (Type 4)

Given:

- Same circuit, but a silicon diode ($V_K = 0.7$ V), $V_{dc} = 2$ V, $R = 2$ k Ω , $f = 60$ Hz

Find:

- Peak value of v_i , and the same three sketches

Formula:

$$V_{dc} = 0.318 (V_m - 0.7) \rightarrow V_m = V_{dc}/0.318 + 0.7$$

Solution:

1. $V_m - 0.7 = 2 / 0.318 = 6.29$ V
2. $V_m = 6.29 + 0.7 = \mathbf{6.99\text{ V} \approx 7.0\text{ V}}$
3. Peak output voltage = $V_m - 0.7 = \mathbf{6.29\text{ V}}$
4. Peak current = $6.29 / 2\text{ k}\Omega = \mathbf{3.14\text{ mA}}$

Sketches — what changes from Q22:

Waveform	Positive half-cycle	Negative half-cycle
v_i	up to +6.99 V	down to -6.99 V
v_d	held at +0.7 V (not 0)	follows v_i to -6.99 V
i_d	half-sine, peak 3.14 mA, but conducts only while $v_i > 0.7$ V	0 mA

Answer:

- Peak value of v_i = **6.99 V**
- v_d = 0.7 V when conducting, and reaches -6.99 V when off
- i_d peak = 3.14 mA; conduction is slightly **less than** a full half-cycle because the diode waits for v_i to exceed 0.7 V

Question 24 — Fig. 2.169 (10 k Ω load added)

Problem Type: Half-wave rectifier with a parallel load (Type 4)

Circuit values read from Fig. 2.169: v_i → ideal diode → output node, with 2 k Ω and $R_L = 10$ k Ω **both** to ground

Given:

- Repeat Q22 (ideal diode, $V_m = 6.29$ V, $f = 60$ Hz), now with $R_L = 10$ k Ω in parallel with the 2 k Ω

Find:

- Sketch v_L and i_L

Formula:

$$R_{\text{parallel}} = (2k \times 10k) / (2k + 10k)$$

$$\text{Diode ON} \rightarrow v_L = v_i ; i_L = v_L / R_L$$

Solution:

1. Parallel resistance = $(2 \times 10) / (2 + 10) \text{ k}\Omega = 20/12 = \mathbf{1.667 \text{ k}\Omega}$
2. The ideal diode connects v_i straight to the output node with no series resistor, so during the positive half-cycle the output simply follows the input:
 $v_{L(\text{peak})} = V_m = \mathbf{6.29 \text{ V}}$
3. Negative half-cycle: diode off $\rightarrow v_L = \mathbf{0 \text{ V}}$
4. Load current: $i_{L(\text{peak})} = 6.29 / 10 \text{ k}\Omega = \mathbf{0.629 \text{ mA}}$
5. Total diode current: $i_{d(\text{peak})} = 6.29 / 1.667 \text{ k}\Omega = \mathbf{3.77 \text{ mA}}$
6. DC level across the load: $V_{L(\text{dc})} = 0.318 \times 6.29 = \mathbf{2 \text{ V}}$ (unchanged)
 $I_{L(\text{dc})} = 2 / 10 \text{ k}\Omega = \mathbf{0.2 \text{ mA}}$

Sketches — what to draw:

Waveform	Positive half-cycle	Negative half-cycle
v_L	half-sine up to 6.29 V	0 V
i_L	half-sine up to 0.629 mA	0 mA

Answer:

- v_L = half-wave rectified sine, peak **6.29 V**, DC level **2 V**
- i_L = half-wave rectified, peak **0.629 mA**, DC level **0.2 mA**
- Adding the load does not change the output **voltage** shape (the ideal diode has no series resistance); it only increases the total current the diode must carry, from 3.14 mA to **3.77 mA**.

Full-Wave Rectification

Question 29 — Fig. 2.173

Problem Type: Full-wave bridge rectifier (Type 5)

Circuit values read from Fig. 2.173: standard four-diode bridge, **ideal diodes**, v_i peak = 100 V, $R_L = 2.2 \text{ k}\Omega$

Given:

- Bridge rectifier, ideal diodes, $V_m = 100 \text{ V}$, $R_L = 2.2 \text{ k}\Omega$

Find:

- v_o , the PIV rating of each diode, and the maximum current through each diode

Formula:

$$V_{o(\text{peak})} = V_m - 2V_K = V_m \quad (\text{ideal})$$

$$V_{dc} = 0.636 V_{o(\text{peak})}$$

$$\text{PIV}(\text{bridge}) = V_m$$

$$I_{\text{max}} = V_{o(\text{peak})} / R_L$$

Solution:

1. Diodes are ideal, so no drops are subtracted:
 v_o = full-wave rectified sine with peak **100 V**
2. DC level: $V_{dc} = 0.636 \times 100 = \mathbf{63.6 \text{ V}}$
3. PIV: when a diode is off in a bridge, it must block the full peak:
 $\text{PIV} = V_m = \mathbf{100 \text{ V}}$

4. Maximum diode current = peak load current (each diode carries the full load current during its half-cycle):

$$I_{\max} = 100 / 2.2 \text{ k}\Omega = \mathbf{45.45 \text{ mA}}$$

Answer:

- v_o = full-wave rectified sine, peak **100 V**, $V_{dc} = \mathbf{63.6 \text{ V}}$
- PIV of each diode = **100 V**
- Maximum current through each diode = **45.45 mA**

Question 30 — Fig. 2.174

Problem Type: Bridge with two arms replaced by resistors (Type 5, variant)

Circuit values read from Fig. 2.174: a bridge whose **top two arms are ideal diodes** and whose **bottom two arms are 2.2 k Ω resistors**; output taken across a 2.2 k Ω load to ground; v_i peak = 100 V

Given:

- $V_m = 100 \text{ V}$, ideal diodes, all three resistors = 2.2 k Ω

Find:

- Sketch v_o and determine the available DC voltage

Formula:

Analyse each half-cycle separately with the conducting diode replaced by a short.

Solution:

1. **Positive half-cycle.** The diode from the top node to the output node conducts (ideal $\rightarrow$ a short); the other diode is reverse-biased and off.
2. Label the output node R and the source's lower node B. With the diode shorted, the source voltage appears directly between R and B.
From R there are two return paths to B: the 2.2 k Ω bridge arm directly, and the 2.2 k Ω load to ground then the other 2.2 k Ω arm up to B.
3. Writing KCL at both nodes and using the fact that all three resistors are equal gives $V_R = -V_B$, and since $V_R - V_B = v_i$:
 $2V_R = v_i \rightarrow v_o = v_i/2$
4. **Negative half-cycle.** Now the other diode conducts, tying the top node to ground, and the analysis mirrors itself:
 $v_o = |v_i|/2$
5. So the output is a **full-wave rectified sine of half the input amplitude**:
 $v_{o(\text{peak})} = 100/2 = \mathbf{50 \text{ V}}$
6. $V_{dc} = 0.636 \times 50 = \mathbf{31.8 \text{ V}}$

Sketch — what to draw:

- A full-wave rectified sine (both humps positive), peak **50 V**, twice the input frequency.

Answer:

- v_o = full-wave rectified sine with peak **50 V**
- Available DC voltage = **31.8 V**

The resistors split the source voltage in half, so this circuit delivers half the DC output of the true bridge in Q29.

Clippers

Your document lists the "Clippers" heading with the note "*how understand its clippers circuit*" and no typed question text — the actual questions are inside the images. They are **Problem 33 (Fig. 2.177)** and **Problem 35 (Fig. 2.179)**, both solved below.

Question 33 — Fig. 2.177

Problem Type: Series clipper, square-wave input (Type 6)

Circuit values read from Fig. 2.177: input is a square wave switching between **+12 V** and **-12 V**

- **(a)** $v_i \rightarrow$ Si diode (pointing right) $\rightarrow 2.2 \text{ k}\Omega \rightarrow$ node $v_o \rightarrow 1.8 \text{ k}\Omega$ to ground
- **(b)** $v_i \rightarrow$ Si diode (pointing right) $\rightarrow 4 \text{ V}$ battery (**- on the left, + on the right**) $\rightarrow$ node $v_o \rightarrow 10 \text{ k}\Omega$ to ground

Given / Find:

- v_o for each network, at both input levels

Formula:

Diode ON $\rightarrow I = (v_i - V_K \pm V_{\text{battery}}) / \Sigma R$, v_o from Ohm's law
 Diode OFF $\rightarrow I = 0 \rightarrow v_o = 0$ (series clipper, no current in the output resistor)

Solution — (a):

1. **When $v_i = +12 \text{ V}$:** the diode is forward-biased.
 $I = (12 - 0.7) / (2.2 \text{ k}\Omega + 1.8 \text{ k}\Omega) = 11.3 / 4000 = 2.825 \text{ mA}$
 $v_o = I \times 1.8 \text{ k}\Omega = (2.825 \text{ mA})(1800) = \mathbf{5.09 \text{ V}}$
2. **When $v_i = -12 \text{ V}$:** the anode is negative $\rightarrow$ diode **OFF**.
 No current flows, so there is no drop across the $1.8 \text{ k}\Omega$:
 $v_o = \mathbf{0 \text{ V}}$

Solution — (b):

3. **When $v_i = +12 \text{ V}$:** assume the diode conducts.
 Walking from the input: $V_{\text{cathode}} = 12 - 0.7 = 11.3 \text{ V}$.
 The battery has **-** on the left and **+** on the right, so crossing it **raises** the potential by 4 V :
 $v_o = 11.3 + 4 = \mathbf{15.3 \text{ V}}$
 Check: current $= 15.3 / 10 \text{ k}\Omega = 1.53 \text{ mA}$, positive and in the diode's direction ✓
4. **When $v_i = -12 \text{ V}$:** assume off. With no current, $v_o = 0 \text{ V}$, so the cathode sits at $0 - 4 = -4 \text{ V}$.
 The anode is at -12 V , well below the cathode $\rightarrow$ **reverse-biased**, confirmed off.
 $v_o = \mathbf{0 \text{ V}}$

Answer:

v_i	(a) v_o	(b) v_o
+12 V	5.09 V	15.3 V
-12 V	0 V	0 V

In (b) the battery **adds** to the output because its **+** terminal faces the output node. Reverse the battery and you would get $12 - 0.7 - 4 = 7.3 \text{ V}$ instead.

Question 35 — Fig. 2.179

Problem Type: Parallel (shunt) clipper with a DC source (Type 6)

Circuit values read from Fig. 2.179: input is a **sine wave of $\pm 8 \text{ V}$**

- **(a)** $v_i \rightarrow 1 \text{ k}\Omega \rightarrow$ node v_o ; from v_o down to the bottom rail: a Si diode (anode at the top) in series with a 4 V battery (**+ on top**)
- **(b)** $v_i \rightarrow 2.2 \text{ k}\Omega \rightarrow 3 \text{ V}$ battery (**+ on the left, - on the right**) $\rightarrow$ node v_o ; from v_o a Si diode (anode at the top) down to ground

Given / Find:

- v_o for each network

Formula:

Clipping level = walk from v_o down the diode branch to ground, adding V_K and the battery
 Diode ON $\rightarrow v_o =$ clipping level
 Diode OFF $\rightarrow$ no branch current $\rightarrow v_o$ follows the input path

Solution — (a):

1. Walk down the branch from v_o : through the diode (+0.7 V) then to the battery's + terminal (+4 V).
Clipping level = $4 + 0.7 = 4.7$ V
2. **While $v_i < 4.7$ V:** the diode is off, the branch carries no current, so there is no drop across the 1 k Ω :
 $v_o = v_i$
3. **While $v_i > 4.7$ V:** the diode conducts and holds the node:
 $v_o = 4.7$ V (flat)
4. The negative peak is untouched: v_o reaches **-8 V**.

Sketch (a): a sine wave whose **top is flattened at +4.7 V**, negative peak still **-8 V**.

Solution — (b):

5. **Diode branch:** the diode's anode is at v_o , cathode at ground $\rightarrow$ it conducts when v_o tries to exceed +0.7 V.
Clipping level = 0.7 V
6. **When the diode is off:** no current flows, so no drop across the 2.2 k Ω . The 3 V battery has + on the left and - on the right, so crossing it toward the output **lowers** the potential by 3 V:
 $v_o = v_i - 3$
7. **Turn-on point:** the diode conducts once $v_i - 3 \geq 0.7$, i.e. once $v_i \geq 3.7$ V.
Above that: $v_o = 0.7$ V (flat)
8. At the negative peak: $v_o = -8 - 3 = -11$ V

Sketch (b): a sine wave **shifted down 3 V** and **flattened at +0.7 V** on top; negative peak **-11 V**.

Answer:

- (a) $v_o = v_i$ for $v_i \leq 4.7$ V, clipped flat at **4.7 V** above that; negative peak **-8 V**
- (b) $v_o = v_i - 3$ for $v_i < 3.7$ V, clipped flat at **0.7 V** above that; negative peak **-11 V**

Clampers

As with the clippers, your document's "Clampers" heading has no typed question text — the questions are in the images. They are **Problem 38 (Fig. 2.182)**, **Problem 39 (Fig. 2.183)**, **Problem 40 (Fig. 2.184)** and **Problem 41 (Fig. 2.185)**, all solved below.

Question 38 — Fig. 2.182

Problem Type: Clamper analysis (Type 7)

Circuit values read from Fig. 2.182: input is a **sine wave of ± 120 V**

- (a) series C $\rightarrow$ node v_o , with an **ideal diode whose cathode is at the top** (so it conducts upward, from ground into the node) in parallel with R
- (b) series C $\rightarrow$ node v_o , with an **ideal diode whose anode is at the top** in series with **E = 20 V (+ on top)**, in parallel with R

Given / Find:

- Sketch v_o for each

Formula:

During conduction: $v_o = \text{clamping level}$, and $V_C = v_i - v_o$
 Diode off: $v_o = v_i - V_C$
 Check: $v_o(\text{p-p})$ must equal $v_i(\text{p-p}) = 240$ V

Solution — (a):

1. The diode's cathode faces the output node, so it conducts whenever v_o tries to go **below 0 V**. Ideal $\rightarrow$ clamping level = **0 V**.
2. This happens during the **negative** half-cycle. At the negative peak, $v_i = -120$ V and $v_o = 0$:
 $V_C = v_i - v_o = -120 - 0 = -120$ V
(i.e. the capacitor holds 120 V with its **right** plate positive)
3. During the positive half-cycle the diode is off, and V_C is held:

$$v_o = v_i - V_C = v_i + 120$$

4. At the positive peak: $v_o = 120 + 120 = \mathbf{+240\text{ V}}$

At the negative peak: $v_o = -120 + 120 = \mathbf{0\text{ V}}$

5. **Check:** peak-to-peak = $240 - 0 = 240\text{ V} = \text{input peak-to-peak} \checkmark$

Sketch (a): the input sine shifted **up by 120 V**, swinging from **0 V to +240 V**. This is a **positive clamper**.

Solution – (b):

6. The diode's anode faces the output node and its cathode goes to the + terminal of the 20 V source, so it conducts whenever v_o tries to rise **above 20 V**. Ideal $\rightarrow$ clamping level = **+20 V**.

7. This happens during the **positive** half-cycle. At the positive peak, $v_i = +120\text{ V}$ and $v_o = 20\text{ V}$:

$$V_C = 120 - 20 = \mathbf{+100\text{ V}} \text{ (left plate positive)}$$

8. During the negative half-cycle the diode is off:

$$v_o = v_i - 100$$

9. At the negative peak: $v_o = -120 - 100 = \mathbf{-220\text{ V}}$

10. **Check:** peak-to-peak = $20 - (-220) = 240\text{ V} \checkmark$

Sketch (b): the input sine shifted **down by 100 V**, swinging from **+20 V to -220 V**. This is a **negative clamper with a 20 V reference**.

Answer:

- (a) v_o swings from **0 V to +240 V** ($V_C = 120\text{ V}$)
- (b) v_o swings from **+20 V to -220 V** ($V_C = 100\text{ V}$)

Question 39 — Fig. 2.183

Problem Type: (a),(b) Time-constant check (Type 9); (c) Clamper sketch (Type 7)

Circuit values read from Fig. 2.183: input is a **square wave of $\pm 12\text{ V}$ at $f = 1\text{ kHz}$** ; $C = 0.1\text{ }\mu\text{F}$; a **Si diode with its anode at the top** in series with a **2 V battery (– on top, + at the bottom rail)**; $R = 56\text{ k}\Omega$ in parallel with the output

Given:

- $C = 0.1\text{ }\mu\text{F}$, $R = 56\text{ k}\Omega$, $f = 1\text{ kHz}$, $v_i = \pm 12\text{ V}$ square wave, Si diode

Find:

- (a) 5τ (b) compare with half the period (c) sketch v_o

Formula:

$$\tau = RC ; \quad T = 1/f ; \quad \text{clamping level from the diode branch ; } V_C = v_i - v_o$$

Solution — part (a):

1. $\tau = RC = (56 \times 10^3)(0.1 \times 10^{-6}) = 5.6 \times 10^{-3}\text{ s} = 5.6\text{ ms}$
2. $5\tau = 5 \times 5.6\text{ ms} = \mathbf{28\text{ ms}}$

Solution — part (b):

3. $T = 1/f = 1/1000 = 1\text{ ms}$, so **$T/2 = 0.5\text{ ms}$**
4. Compare: 28 ms vs $0.5\text{ ms} \rightarrow \mathbf{5\tau}$ is **56 times larger than the half period**
5. Because $5\tau \gg T/2$, the capacitor has almost no time to discharge during a half-cycle. Its voltage stays essentially constant, so the clamping action is very good and the output holds a flat top.

Solution — part (c):

6. **Clamping level.** The battery has its – terminal at the top, so the diode's cathode sits at $0 - 2 = -2\text{ V}$. The diode conducts when $v_o \geq -2 + 0.7 = \mathbf{-1.3\text{ V}}$.
7. **During $v_i = +12\text{ V}$:** the diode conducts and holds the node at $v_o = \mathbf{-1.3\text{ V}}$
 $V_C = v_i - v_o = 12 - (-1.3) = \mathbf{13.3\text{ V}}$
8. **During $v_i = -12\text{ V}$:** the diode is off and V_C is held:
 $v_o = -12 - 13.3 = \mathbf{-25.3\text{ V}}$

9. **Check:** peak-to-peak = $-1.3 - (-25.3) = 24 \text{ V} = \text{input peak-to-peak } (12 + 12) \checkmark$

Sketch (c): a square wave alternating between -1.3 V and -25.3 V , with the flat top at -1.3 V lasting 0.5 ms and the bottom at -25.3 V lasting 0.5 ms .

Answer:

- (a) $5\tau = 28 \text{ ms}$
- (b) $T/2 = 0.5 \text{ ms}$; 5τ is **56× larger**, so the capacitor barely discharges and clamping is excellent
- (c) v_o is a square wave between -1.3 V and -25.3 V ($V_C = 13.3 \text{ V}$)

Question 40 — Fig. 2.184

Problem Type: Clamper design, ideal diodes (Type 8)

Waveforms read from Fig. 2.184:

- Input: square wave between **+20 V and -20 V** (40 V peak-to-peak)
- Required output: square wave between **+30 V and -10 V** (40 V peak-to-peak)
- Ideal diodes

Given / Find:

- Design the clamper — determine the DC source and the diode orientation

Formula:

Shift = $v_o - v_i$ (must match at both levels)
 Clamping level = the held level ; $V_{\text{battery}} = \text{clamping level} \mp V_K$
 Design rule: $5RC \gg T/2$

Solution:

1. **Check the swing:** input p-p = $20 - (-20) = 40 \text{ V}$; output p-p = $30 - (-10) = 40 \text{ V} \checkmark$
 A clamper can do this (it shifts without changing the swing).
2. **Find the shift:**
 $+20 \text{ V} \rightarrow +30 \text{ V}$: shift = **+10 V**
 $-20 \text{ V} \rightarrow -10 \text{ V}$: shift = **+10 V** ✓ consistent
 So we need a **positive shift of 10 V**.
3. **Identify the clamped level.** The output's lower level, **-10 V**, is the one the diode must hold.
4. **Capacitor voltage.** During the negative half ($v_i = -20 \text{ V}$), the diode conducts and $v_o = -10 \text{ V}$:
 $V_C = v_i - v_o = -20 - (-10) = \textbf{-10 V}$ (right plate positive)
5. **Verify the other half.** With the diode off: $v_o = v_i - V_C = 20 - (-10) = \textbf{+30 V} \checkmark$
6. **Circuit design:**
 - A capacitor **C in series** with the input.
 - A shunt branch from the output node to ground: an **ideal diode with its cathode facing the output node** (so it conducts upward, into the node, when v_o falls) in series with a **10 V DC source whose negative terminal faces the diode** (so the diode's anode sits at -10 V).
 - Ideal diode $\rightarrow V_K = 0$, so the clamping level = the source voltage = **-10 V** ✓
 - A resistor **R in parallel** with the diode branch.
7. **Component sizing:** the frequency is **not given** in Fig. 2.184, so exact R and C values cannot be pinned down. State the design condition instead:
 $5RC \gg T/2$, where $T = 1/f$ of the input. (For example, if f were 1 kHz , then $T/2 = 0.5 \text{ ms}$ and $R = 100 \text{ k}\Omega$ with $C = 1 \text{ }\mu\text{F}$ would give $5RC = 500 \text{ ms}$, comfortably satisfying the rule.)

Answer:

- **Series capacitor C**, plus a shunt branch containing an **ideal diode (cathode toward the output node)** in series with a **10 V DC source (negative terminal toward the diode)**, and **R in parallel**.
- This clamps the lower level at -10 V and produces $V_C = 10 \text{ V}$, giving an output of $+30 \text{ V} / -10 \text{ V}$.
- Choose R and C so that **$5RC \gg T/2$** . The frequency is not stated in the figure, so no numerical R and C can be

determined from the given information.

Question 41 — Fig. 2.185

Problem Type: Clamper design, silicon diodes (Type 8)

Waveforms read from Fig. 2.185:

- Input: square wave between **+10 V and -10 V** (20 V peak-to-peak)
- Required output: square wave between **+2.7 V and -17.3 V** (20 V peak-to-peak)
- Silicon diodes ($V_K = 0.7 \text{ V}$)

Given / Find:

- Design the clamper

Formula:

$$\text{Shift} = v_o - v_i ; \quad \text{clamping level} = V_{\text{battery}} + V_K \quad (\text{for a diode with its anode at the output node})$$

Solution:

1. **Check the swing:** input p-p = 20 V; output p-p = $2.7 - (-17.3) = 20 \text{ V}$ ✓
2. **Find the shift:**
 $+10 \text{ V} \rightarrow +2.7 \text{ V}$; shift = **-7.3 V**
 $-10 \text{ V} \rightarrow -17.3 \text{ V}$; shift = **-7.3 V** ✓ consistent
 So we need a **negative shift of 7.3 V**.
3. **Identify the clamped level.** The output's upper level, **+2.7 V**, is the one held flat.
4. **Capacitor voltage.** During the positive half ($v_i = +10 \text{ V}$), the diode conducts and $v_o = +2.7 \text{ V}$:
 $V_C = 10 - 2.7 = \mathbf{+7.3 \text{ V}}$ (left plate positive)
5. **Verify the other half.** Diode off: $v_o = v_i - V_C = -10 - 7.3 = \mathbf{-17.3 \text{ V}}$ ✓
6. **Find the DC source.** The diode's anode is at the output node and its cathode goes to the source. When conducting:
 $v_o = V_{\text{source}} + V_K$
 $2.7 = V_{\text{source}} + 0.7 \rightarrow \mathbf{V_{\text{source}} = 2 \text{ V}}$
7. **Circuit design:**
 - A capacitor **C in series** with the input.
 - A shunt branch from the output node to ground: a **silicon diode with its anode facing the output node**, in series with a **2 V DC source whose positive terminal faces the diode**.
 - A resistor **R in parallel**.
8. **Component sizing:** again the frequency is **not given** in Fig. 2.185, so state the condition $\mathbf{5RC \gg T/2}$ rather than inventing numbers.

Answer:

- **Series capacitor C**, plus a shunt branch containing a **silicon diode (anode toward the output node)** in series with a **2 V DC source (positive terminal toward the diode)**, and **R in parallel**.
- Clamping level = $2 + 0.7 = \mathbf{2.7 \text{ V}}$ ✓, giving $V_C = 7.3 \text{ V}$ and an output of $+2.7 \text{ V} / -17.3 \text{ V}$.
- Choose R and C so that $\mathbf{5RC \gg T/2}$. The frequency is not stated in the figure.

Compare Q40 and Q41: in Q40 the diode's **cathode** faces the output (clamping the bottom, shifting up); in Q41 the **anode** faces the output (clamping the top, shifting down). That single reversal is what sets the direction of the shift.

Zener Diodes

Question 42 — Fig. 2.186

Problem Type: Zener regulator with variable R_L (Type 10)

Circuit values read from Fig. 2.186: $V_i = 20 \text{ V}$, $R_S = 220 \Omega$, $V_Z = 10 \text{ V}$, $P_{Z\text{max}} = 400 \text{ mW}$, R_L variable

Given:

- $V_i = 20 \text{ V}$, $R_S = 220 \text{ } \Omega$, $V_Z = 10 \text{ V}$, $P_{Z\max} = 400 \text{ mW}$

Find:

- **(a)** V_L , I_L , I_Z , I_R for $R_L = 180 \text{ } \Omega$
- **(b)** the same for $R_L = 470 \text{ } \Omega$
- **(c)** R_L for maximum Zener power
- **(d)** minimum R_L to keep the Zener on

Formula:

Test: $V = V_i R_L / (R_S + R_L)$ compare with V_Z
 Zener ON: $V_L = V_Z$; $I_R = (V_i - V_Z) / R_S$; $I_L = V_Z / R_L$; $I_Z = I_R - I_L$
 $I_{Z\max} = P_{Z\max} / V_Z$; $R_{L\min} = R_S V_Z / (V_i - V_Z)$

Solution — part (a), $R_L = 180 \text{ } \Omega$:

1. **Test:** $V = 20 \times 180 / (220 + 180) = 3600 / 400 = 9 \text{ V}$
2. $9 \text{ V} < V_Z = 10 \text{ V} \rightarrow$ the **Zener is OFF**. It is just a voltage divider.
3. $V_L = 9 \text{ V}$
4. $I_L = 9 / 180 = 50 \text{ mA}$
5. $I_Z = 0 \text{ mA}$
6. $I_R = I_L = 50 \text{ mA}$

Solution — part (b), $R_L = 470 \text{ } \Omega$:

7. **Test:** $V = 20 \times 470 / (220 + 470) = 9400 / 690 = 13.62 \text{ V}$
8. $13.62 \text{ V} > 10 \text{ V} \rightarrow$ the **Zener is ON**.
9. $V_L = 10 \text{ V}$
10. $I_R = (20 - 10) / 220 = 10 / 220 = 45.45 \text{ mA}$
11. $I_L = 10 / 470 = 21.28 \text{ mA}$
12. $I_Z = 45.45 - 21.28 = 24.17 \text{ mA}$
13. Power check: $P_Z = 10 \times 24.17 \text{ mA} = 242 \text{ mW} < 400 \text{ mW} \checkmark$ safe

Solution — part (c), R_L for maximum Zener power:

14. $I_{Z\max} = P_{Z\max} / V_Z = 400 \text{ mW} / 10 \text{ V} = 40 \text{ mA}$
15. With the Zener on, I_R is fixed at 45.45 mA.
16. $I_L = I_R - I_{Z\max} = 45.45 - 40 = 5.45 \text{ mA}$
17. $R_L = V_Z / I_L = 10 / 5.45 \text{ mA} = 1.83 \text{ k}\Omega$

Solution — part (d), minimum R_L :

18. $R_{L\min} = R_S V_Z / (V_i - V_Z) = (220 \times 10) / (20 - 10) = 2200 / 10 = 220 \text{ } \Omega$

Answer:

- **(a)** $V_L = 9 \text{ V}$, $I_L = 50 \text{ mA}$, $I_Z = 0 \text{ mA}$, $I_R = 50 \text{ mA}$ (**Zener off**)
- **(b)** $V_L = 10 \text{ V}$, $I_L = 21.28 \text{ mA}$, $I_Z = 24.17 \text{ mA}$, $I_R = 45.45 \text{ mA}$ (**Zener on**)
- **(c)** $R_L = 1.83 \text{ k}\Omega$ gives maximum Zener power
- **(d)** $R_{L\min} = 220 \text{ } \Omega$

Part (a) is the trap: $R_L = 180 \text{ } \Omega$ is **below** the $220 \text{ } \Omega$ minimum found in part (d), which is exactly why the Zener cannot turn on.

Question 43 — Fig. 2.187

Problem Type: Zener design (Type 12)

Circuit values read from Fig. 2.187: $V_i = 16 \text{ V}$, R_S unknown, Zener with V_Z unknown, R_L variable

Given:

- $V_i = 16 \text{ V}$, required $V_L = 12 \text{ V}$, load current I_L varies from **0 mA to 200 mA**

Find:

- (a) R_S and V_Z (b) P_{Zmax}

Formula:

$$\begin{aligned} V_Z &= V_L \\ R_S &= (V_i - V_Z)/I_{L(max)} \\ \text{Worst case for the Zener is } I_L &= 0 \rightarrow I_Z = I_R \\ P_{Zmax} &= V_Z \times I_R \end{aligned}$$

Solution — part (a):

1. The Zener sets the output, so $V_Z = V_L = 12 \text{ V}$
2. Size R_S at the **heaviest** load, $I_L = 200 \text{ mA}$ (this is when the Zener has the least current to spare, so set $I_Z \approx 0$):
 $I_R = I_{L(max)} = 200 \text{ mA}$
3. Voltage across $R_S = 16 - 12 = 4 \text{ V}$
 $R_S = 4 / 200 \text{ mA} = 20 \Omega$

Solution — part (b):

4. The Zener's worst case is the **lightest** load, $I_L = 0 \text{ mA}$. Then it must absorb the entire I_R :
 $I_Z = I_R = (16 - 12)/20 = 200 \text{ mA}$
5. $P_{Zmax} = V_Z \times I_Z = 12 \times 0.2 = 2.4 \text{ W}$

Answer:

- (a) $R_S = 20 \Omega$, $V_Z = 12 \text{ V}$
- (b) $P_{Zmax} = 2.4 \text{ W}$ (the Zener must be rated for at least this)

Note the opposite extremes: **R_S is sized at maximum load**, but the **Zener's power rating is set by minimum load**.

Question 44 — Fig. 2.188

Problem Type: Zener regulator $\rightarrow$ range of V_i (Type 11)

Circuit values read from Fig. 2.188: $R_S = 91 \Omega$, $V_Z = 8 \text{ V}$, $P_{Zmax} = 400 \text{ mW}$, $R_L = 0.22 \text{ k}\Omega = 220 \Omega$

Given:

- $R_S = 91 \Omega$, $V_Z = 8 \text{ V}$, $P_{Zmax} = 400 \text{ mW}$, $R_L = 220 \Omega$ (fixed)

Find:

- The range of V_i that maintains $V_L = 8 \text{ V}$ without exceeding the Zener's power rating

Formula:

$$\begin{aligned} I_L &= V_Z/R_L \quad (\text{fixed}) \\ I_{Zmax} &= P_{Zmax}/V_Z \\ V_{i(min)} &= V_Z + I_L R_S \quad (I_Z = 0) \\ V_{i(max)} &= V_Z + (I_L + I_{Zmax}) R_S \quad (I_Z = I_{Zmax}) \end{aligned}$$

Solution:

1. R_L is fixed and V_L is held at 8 V , so the load current never changes:
 $I_L = 8 / 220 = 36.36 \text{ mA}$
2. $I_{Zmax} = P_{Zmax}/V_Z = 400 \text{ mW} / 8 \text{ V} = 50 \text{ mA}$
3. **Minimum V_i** — the Zener is just barely on, so $I_Z = 0$ and $I_R = I_L = 36.36 \text{ mA}$:
 $V_{i(min)} = 8 + (36.36 \text{ mA})(91 \Omega) = 8 + 3.31 = 11.31 \text{ V}$
4. **Maximum V_i** — the Zener is at its power limit, so $I_R = I_L + I_{Zmax} = 36.36 + 50 = 86.36 \text{ mA}$:
 $V_{i(max)} = 8 + (86.36 \text{ mA})(91 \Omega) = 8 + 7.86 = 15.86 \text{ V}$

Answer:

- $11.31 \text{ V} \leq V_i \leq 15.86 \text{ V}$

- Below 11.31 V the Zener drops out of regulation and V_L falls below 8 V; above 15.86 V the Zener exceeds its 400 mW rating.

FINAL EXAM QUICK REVISION

1. Chapter → Problem Type

Chapter	Problem Type	How to Recognize	Main Formula
4	Fixed bias — Q-point	One R_B , emitter grounded	$I_B = (V_{CC} - 0.7)/R_B$
4	Emitter bias — Q-point	R_B to V_{CC} plus R_E	$I_B = (V_{CC} - 0.7)/[R_B + (\beta + 1)R_E]$
4	Voltage divider — exact	Two base resistors R_1, R_2	R_{Th}, E_{Th} , then $I_B = (E_{Th} - 0.7)/[R_{Th} + (\beta + 1)R_E]$
4	Voltage divider — approx.	$\beta R_E \geq 10R_2$ holds	$V_B = V_{CC}R_2/(R_1 + R_2)$
4	Collector feedback	R_F from base to collector	$I_B = (V_{CC} - 0.7)/[R_F + \beta(R_C + R_E)]$
4	Emitter follower	No R_C , output at emitter	Same as emitter bias; $V_C = V_{CC}$
4	Common base	Base grounded / fixed V_B	$V_E = V_B - 0.7$, then $I_E = \Delta V/R_E$
4	Saturation current	Word " I_{Csat} "	$I_{Csat} = V_{CC}/(R_C + R_E)$
4	Reverse problem	Letters on resistors, numbers on nodes	Rearrange; start where V and I are both known
4	Design	"Design...", "Use standard values"	$R = V_{across}/I_{through}$, then round values
4	Current source	"Calculate the current I "	Find $V_B \rightarrow V_E \rightarrow I_E \rightarrow I_C$
1	Thermal voltage	Temperature in $^{\circ}\text{C}$ given	$V_T = (8.62 \times 10^{-5})(T_C + 273)$
1	Diode current	I_s , n , positive V_D	$I_D = I_s(e^{V_D/nV_T} - 1)$
1	Reverse bias	Negative V_D	$I_D = -I_s$
1	Find I_s	"find I_s "	$I_s = I_D/(e^{V_D/nV_T} - 1)$
1	Find V_D	"find the applied voltage"	$V_D = nV_T \ln(I_D/I_s + 1)$
2	Series diode	One loop, diodes + resistors	Check ON/OFF, then $I = (\Sigma V - \Sigma V_K)/\Sigma R$
2	Opposing diodes	Arrows facing each other	Open circuit $\rightarrow I = 0$
2	Parallel diode branches	Two diode branches meeting	KCL at the output node
2	Half-wave rectifier	One diode + sine	$V_{dc} = 0.318(V_m - V_K)$
2	Bridge rectifier	Four diodes in a diamond	$V_{dc} = 0.636V_m$, $\text{PIV} = V_m$
2	Clipper	Diode across output, no capacitor	Clipping level = $V_{batt} + V_K$
2	Clamper	Capacitor in series	$V_C = v_i - v_o$, then $v_o = v_i - V_C$
2	Clamper design	Two waveforms shown	Find the shift and the clamped level
2	Zener, variable R_L	Zener across a load	Test $V = V_i R_L/(R_S + R_L)$ first
2	Zener, range of V_i	R_L fixed, "range of V_i "	$I_Z = 0$ and $I_Z = I_{Zmax}$
2	Zener design	"Design... maintain V_L at..."	R_S at max load; P_Z at min load

2. Formula Checklist

Tick these off before the exam. If you can write all of these from memory, you can do every question in this guide.

Chapter 4 — must know

- ☐ $I_C = \beta I_B$, $I_E = (\beta + 1)I_B$, $\alpha = \beta/(\beta + 1)$
- ☐ $V_B = V_E + 0.7$, $V_{CE} = V_C - V_E$
- ☐ Fixed bias: $I_B = (V_{CC} - 0.7)/R_B$
- ☐ Emitter bias: $I_B = (V_{CC} - 0.7)/[R_B + (\beta + 1)R_E]$
- ☐ Thévenin: $R_{Th} = R_1 \parallel R_2$, $E_{Th} = V_{CC}R_2/(R_1 + R_2)$
- ☐ Eq. (4.33): $\beta R_E \geq 10R_2$
- ☐ Collector feedback: $I_B = (V_{CC} - 0.7)/[R_F + \beta(R_C + R_E)]$
- ☐ $I_{Csat} = V_{CC}/(R_C + R_E)$
- ☐ Divider reverse: $I_1 = I_2 + I_B$

Chapter 1 — must know

- ☐ $V_T = (8.62 \times 10^{-5})(T_C + 273)$
- ☐ $I_D = I_S(e^{V_D/(nV_T)} - 1)$
- ☐ $V_D = nV_T \ln(I_D/I_S + 1)$
- ☐ Reverse bias $\rightarrow I_D = -I_S$

Chapter 2 — must know

- ☐ V_K : Ideal 0, Ge 0.3, Si 0.7, GaAs 1.2
- ☐ $I = (\Sigma V - \Sigma V_K)/\Sigma R$
- ☐ Half-wave: $V_{dc} = 0.318(V_m - V_K)$
- ☐ Full-wave: $V_{dc} = 0.636V_m$, $PIV = V_m$
- ☐ Clipping level = $V_{battery} + V_K$
- ☐ Clamper: $V_C = v_i - v_o$ during conduction; then $v_o = v_i - V_C$
- ☐ Zener test: $V = V_i R_L/(R_S + R_L)$
- ☐ $I_Z = I_R - I_L$, $I_{Zmax} = P_{Zmax}/V_Z$
- ☐ $R_{Lmin} = R_S V_Z/(V_i - V_Z)$

3. “When I See This $\rightarrow$ Do This”

Chapter 4

- If you see **only one resistor at the base and the emitter grounded** $\rightarrow$ fixed bias $\rightarrow I_B = (V_{CC} - 0.7)/R_B$
- If you see **R_E anywhere** $\rightarrow$ the base-loop denominator becomes $R_B + (\beta + 1)R_E$
- If you see **two resistors at the base** $\rightarrow$ find R_{Th} and E_{Th} first $\rightarrow$ then treat it as emitter bias
- If the question says “**approximate approach**” $\rightarrow$ first check $\beta R_E \geq 10R_2$, then read V_B straight off the divider
- If the base resistor **ends at the collector** $\rightarrow$ use $\beta(R_C + R_E)$, not $(\beta + 1)R_E$
- If you see a **capacitor** in a DC bias problem $\rightarrow$ treat it as an **open circuit**; series resistors around it simply add
- If you see **no R_C** $\rightarrow V_C = V_{CC}$ immediately
- If the **base is grounded** $\rightarrow V_E = -0.7$ V immediately, then use Ohm’s law
- If you see **two supplies (+V and -V)** $\rightarrow$ the base-loop driving voltage is $V_{top} + |V_{bottom}| - 0.7$
- If the question asks for **I_{Csat}** $\rightarrow$ just $V_{CC} \div (R_C + R_E)$; ignore β entirely
- If **resistors are labelled with letters** and node voltages are given $\rightarrow$ reverse problem $\rightarrow$ start at a resistor where you know both V and I
- If you are finding **R_1 in a divider** $\rightarrow$ remember $I_1 = I_2 + I_B$
- If the question says “**calculate the current I**” with a load in the collector $\rightarrow$ current source $\rightarrow$ the load value is irrelevant
- If a **Zener sets the base voltage** $\rightarrow V_B = V_{rail} + V_Z$ directly, no KVL needed

Chapter 1

- If a **temperature in °C** appears $\rightarrow$ add 273 first, always
- If **n** is given $\rightarrow$ multiply it by V_T , never by V_D
- If **V_D is negative** $\rightarrow$ the answer is $-I_S$; stop, no calculator needed
- If you must **find V_D** $\rightarrow$ use **ln**, never $\log_{10}$

Chapter 2

- If a diode's **anode is more negative** than its cathode $\rightarrow$ OFF $\rightarrow I = 0 \rightarrow V_o = 0$
- If **two diodes face each other** in one branch $\rightarrow$ that branch is an open circuit
- If **two diode branches meet** at a node $\rightarrow$ write KCL, then check both currents are positive
- If a **DC level is given** for a rectifier $\rightarrow$ work backwards: $V_m = V_{dc}/0.318$
- If a rectifier diode is **OFF** $\rightarrow$ the whole input appears **across the diode**
- If you see a **capacitor in series with the input** $\rightarrow$ clamper, not clipper
- If it is a **clamper** $\rightarrow$ find V_C during the conducting half, then check the peak-to-peak matches
- If you see a **Zener** $\rightarrow$ do the removal test *before* assuming it regulates
- If **R_L is fixed** and the question asks for a range $\rightarrow$ two cases: $I_Z = 0$ and $I_Z = I_{Zmax}$

Unit conversions — do these first, every time

- $k\Omega \rightarrow \Omega$: multiply by 1000 ($2.2\ k\Omega = 2200\ \Omega$)
- $mA \rightarrow A$: divide by 1000 ($2.5\ mA = 0.0025\ A$)
- $\mu A \rightarrow A$: divide by 1 000 000 ($20\ \mu A = 0.00002\ A$)
- **Shortcut:** volts $\div k\Omega =$ **mA** directly. ($11.3\ V \div 4\ k\Omega = 2.825\ mA$)
- **Shortcut:** $k\Omega \times \mu F =$ **milliseconds**. ($56\ k\Omega \times 0.1\ \mu F = 5.6\ ms$)
- $mW \rightarrow W$: divide by 1000 ($400\ mW = 0.4\ W$)

4. Common Mistakes

Unit and arithmetic slips

1. **Forgetting to convert $k\Omega$ to Ω** before dividing — the single most common source of answers that are off by 1000.
2. **Mixing μA and mA** in the same equation. Convert everything to base units, or use the volts $\div k\Omega =$ mA shortcut consistently.
3. **Adding 273 to get Kelvin** — skipping this makes every Chapter 1 answer wrong.
4. **Using log instead of ln** when finding V_D .

Chapter 4 mistakes

5. **Using R_B alone when R_E is present.** The denominator must be $R_B + (\beta+1)R_E$.
6. **Using $(\beta+1)$ in the collector-feedback formula.** That one uses β , and it multiplies $(R_C + R_E)$, not just R_E .
7. **Forgetting I_B when finding R_1 .** The current through R_1 is $I_2 + I_B$ (see Q18).
8. **Using I_C where I_E belongs.** $V_E = I_E R_E$, but $V_C = V_{CC} - I_C R_C$.
9. **Writing $V_{CE} = V_C$** when there is an R_E . That shortcut only works if the emitter is grounded.
10. **Using the approximate divider method without checking Eq. (4.33)** — the exam often awards marks specifically for stating the check.
11. **Treating capacitors as shorts in a DC bias problem.** For DC they are **open**.
12. **Losing the sign of V_E** in split-supply circuits. In Q34 the emitter sits at $-2.55\ V$, below ground.
13. **Putting β into I_{Csat} .** Saturation current has nothing to do with β .
14. **Forgetting to round to a standard value** in design questions when the question explicitly asks for it.

Chapter 2 mistakes

15. **Assuming a diode conducts without testing.** Always check the direction against the source polarity first — several questions here (Q5a, Q7b, Q9b) have the diode off.
16. **Misreading which battery terminal is which.** A “—” at the top of a battery makes the top node **negative**. This flips the answer entirely in Q5(a).
17. **Missing back-to-back diodes.** Two arrows facing each other mean the branch is dead — check before doing any algebra.
18. **Drawing v_d as zero during the OFF half-cycle.** The entire input appears across the diode, so v_d swings to $-V_m$.
19. **Confusing 0.318 and 0.636.** Half-wave uses 0.318, full-wave uses 0.636.
20. **Subtracting one diode drop in a bridge.** Two diodes conduct at once, so subtract $2V_K$.
21. **Confusing a clipper with a clamper.** A **capacitor in series** means clamper. Clipper cuts the waveform; clamper shifts it.

22. **Not checking the clamper peak-to-peak.** The output swing must equal the input swing — this catches nearly every sign error.
23. **Getting the sign of the battery wrong in a clipper.** Walk from the output node down through the branch and add the drops in order.
24. **Assuming the Zener is on.** Always compute $V = V_i R_L / (R_S + R_L)$ first. In Q42(a) it is off.
25. **Forgetting that I_R stays constant** while a Zener regulates — only I_L and I_Z trade current between them.
26. **Sizing R_S at minimum load in a Zener design.** R_S is set by the **maximum** load; the Zener's power rating is set by the **minimum** load.

5. Exam-Day Strategy

1. **Scan the whole paper first** and mark the type of each question in the margin. Types 7 (I_{Csat}) and 3 (reverse-bias diode) take under a minute — do them first for guaranteed marks.
2. **Write the formula before substituting.** Most papers award a mark for the correct formula even if the arithmetic slips.
3. **Convert all units in one line** before you start calculating.
4. **Sanity-check every answer:**
 - I_B should be in **μA** ; I_C in **mA**
 - V_{CE} must be between 0 and V_{CC} — if it is negative or larger than V_{CC} , you have made an error
 - I_{CQ} must be **less than** I_{Csat}
 - V_{BC} should be **negative** in the active region
 - A diode's forward voltage should land near 0.3–0.8 V
 - A clamper output must have the **same peak-to-peak swing** as its input
5. **Show the ON/OFF reasoning** for every diode. Even one line — “anode at -12 V, cathode at ground $\rightarrow$ reverse-biased $\rightarrow I = 0$ ” — usually earns the mark.
6. **If a value seems missing**, say so explicitly and state the method. Never invent a number.